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PHYSICS

1) An electric dipole with dipole moment $\vec{p} = (3\hat{i} + 4\hat{j}) \times 10^{-30}$ C-m is placed in an electric field $\vec{E} = 4000\hat{i}$(N/C). An external agent turns the dipole slowly until its electric dipole moment becomes $\left(-4\hat{i} + 3\hat{j}\right) \times 10^{-30}$ C-m. The work done by the external agent is equal to :–

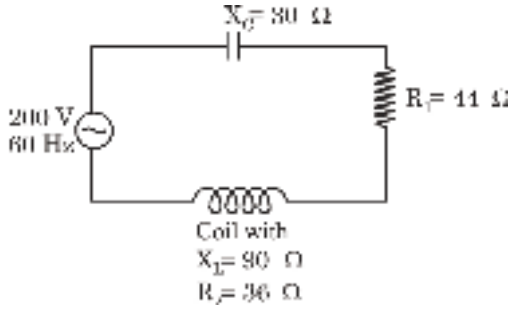
(1) 4×10^{-28} J

(2) -4×10^{-28} J

(3) 2.8×10^{-26} J

(4) -2.8×10^{-26} J

2) As given in the figure, a series circuit connected across a 200 V, 60 Hz line consists of a capacitor of capacitive reactance 30 Ω, a non-inductive resistor of 44 Ω, and a coil of inductive reactance 90Ω and resistance 36Ω. The power dissipated in the coil is :



(1) 320 W

(2) 176 W

(3) 144 W

(4) 0 W

3) The momentum of a photon is 2×10^{-16} g-cm/s. Its energy is :-

(1) 0.61×10^{-26} erg

(2) 2.0×10^{-26} erg

(3) 6×10^{-6} erg

(4) 6×10^{-8} erg

4) The total magnification produced by a compound microscope is 20, while that produced by the eye piece alone is 5. When the microscope is focussed on a certain object, the distance between objective and eye piece is 14 cm. Find the focal length of objective and eye piece, if final image is formed at infinity. (Least distance of distinct vision is 20 cm) :-

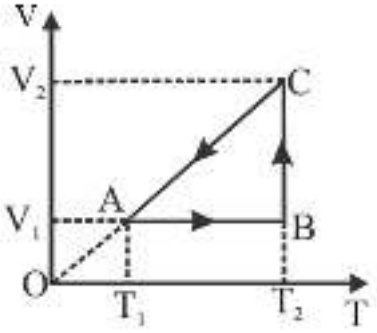
(1) 1.8 cm, 5 cm

(2) 3.33 cm, 4 cm

(3) 9 cm, 5 cm

(4) 2 cm, 4 cm

5) A cyclic process for 1 mole of an ideal gas is shown in figure in the V–T, diagram. The work done in AB, BC and CA respectively :-



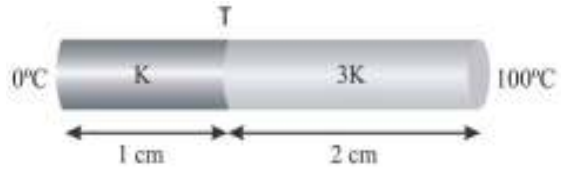
(1) $0, RT_2\ell n\left(\frac{V_1}{V_2}\right), R(T_1 - T_2)$

(2) $R(T_1 - T_2), 0, RT_1\ell n\frac{V_1}{V_2}$

(3) $0, RT_2\ell n\left(\frac{V_2}{V_1}\right), R(T_1 - T_2)$

(4) $0, RT_2\ell n\left(\frac{V_2}{V_1}\right), R(T_2 - T_1)$

6) Two bars of thermal conductivities K and 3 K and lengths 1 cm and 2 cm respectively have same cross-sectional area. They are joined lengths wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature of the interface (T) is :-



(1) 50°C

(2) $\frac{100}{3}$ °C

(3) 60°C

(4) $\frac{200}{3}$ °C

7) Capacitance of a capacitor becomes $\frac{7}{6}$ times of its original value, if a dielectric slab of thickness t = $\frac{2d}{3}$ is inserted between plates. The dielectric constant of slab is (d is separation between the plates) :-

(1) $\frac{14}{11}$

(2) $\frac{11}{14}$

(3) $\frac{7}{11}$

(4) $\frac{11}{7}$

8) A doubly ionized lithium atom is hydrogen like with atomic number 3. Find the wavelength of radiation required to excite electron in Li^{2+} from first to third Bohr's orbit. (hc = 1240 eVmm)

(1) 11.4 nm

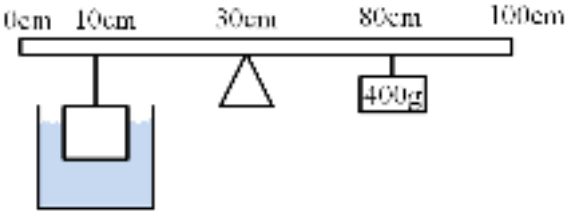
(2) 10.7 nm

(3) 12.7 nm

(4) 13.4 nm

9)

A cube of side 10 cm of unknown mass is suspended from 10 cm mark of a uniform massless meter scale, kept on a knife edge placed at 30 cm mark and 400 g mass is suspended at 80 cm mark as shown. Initially the scale is unbalanced, but when the cube is dipped in beaker filled with water such that half the cube is immersed in water, the scale attains balance. The mass of cube is
(Take density of cube greater than that of water and density of water = 1 g/cm^3).



(Scale in balanced condition)

(1) 1100 g

(2) 1200 g

(3) 1500 g

(4) 1800 g

10) A student is given a group of 4 unlabelled components. He obtained specific observations with them listed in Column-II, when connected with a variable d.c. supply. Match the components listed in column-I with their suitable predicted behaviour listed in column-II.

	Column-I		**Column-II**
(A)	Resistor	(P)	Shows open circuit in one direction of current (up to certain voltage) but fixed voltage drop in opposite direction of current.
(B)	Capacitor	(Q)	Voltage across component remains almost constant when current increases

(C)	Diode/LED	(R)	Shows initial low resistance that increases to infinity.
(D)	Zener diode	(S)	No polarity dependence ; V-I graph is a straight line.

(1) A-S, B-R, C-Q, D-P

(2) A-S, B-R, C-P, D-Q

(3) A-S, B-P, C-Q, D-R

(4) A-Q, B-R, C-P, D-S

11) In a meter bridge experiment the null point is obtained at a distance of 75 cm from terminal A (beginning of scale) when known resistance R is connected to left gap near A. When a resistance of 12 Ω is connected parallel to unknown resistance X in right gap near to terminal B, it was found that null point shifted to 80 cm. The value of unknown resistance X is :

(1) 4 Ω

(2) 8 Ω

(3) 10 Ω

(4) 16 Ω

12) Two coherent sources of intensity ratio 1 : 4 produce an interference pattern. The fringe visibility will be

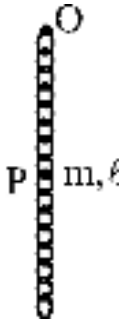
(1) 1

(2) 0.8

(3) 0.4

(4) 0.6

13) A chain of mass m & length is suspended from a point O. Now, the midpoint P of the chain is also taken & slowly put at point O. Find the work done by gravity in this process.



(1) $\frac{3mg\ell}{16}$

(2) $\frac{9mg\ell}{40}$

(3) $\frac{5mg\ell}{16}$

(4) $\frac{7mg\ell}{16}$

14) A man walks in rain at 72 cm/s due east and observes the rain falling vertically. When he stops, rain appears to strikes his back at 37° from the vertical. Find velocity of raindrops relative to the ground.

(1) 100 cm/s

(2) 26 m/s

(3) 120 cm/s

(4) 54 cm/s

15) A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was :-

(1) 510 Hz

(2) 514 Hz

(3) 516 Hz

(4) 508 Hz

16) A ball is thrown on a horizontal surface in such a way that it initially slides with a speed v without rolling. It gradually picks up rotation motion. Find the speed of the ball at which there will be rolling without slipping–

(1) $\frac{2}{7}v_0$

(2) $\frac{2}{5}v_0$

(3) $\frac{5}{7}v_0$

(4) $\frac{3}{5}v_0$

17) Nickel is ferromagnetic at room temperature. At a value of H field of 5×10^5 A/m, nickel saturates. The magnetic field B is 1.25 T. What is the intensity of magnetisation of nickel sample ?

(1) 5×10^5 A/m

(2) $+3.744 \times 10^5$ A/m

(3) 9.31×10^4 A/m

(4) 15×10^5 A/m

18) A particle of mass m is connected with inextensible thread of length ℓ is whirled in vertical circular path. The ratio of maximum tension to minimum tension is 1.5. How much speed at lowest point will be required to achieve that ratio :

(1) $u = \sqrt{5g\ell}$

(2) $u = \sqrt{15g\ell}$

(3) $u = \sqrt{17g\ell}$

(4) $u = \sqrt{18g\ell}$

19) A rocket of mass 6000 kg is set for vertical firing. If the relative exhaust speed be 1 km/sec, how much gas must be ejected per second to give the rocket an upward acceleration of 20 m/s^2 ? [(g = 10 m/s^2)]

(1) 45 kg/sec

(2) 90 kg/sec

(3) 120 kg/sec

(4) 180 kg/sec

20) A particle of mass m and charge q, accelerated by a potential difference V enters a region of a uniform transverse magnetic field B in a direction perpendicular to boundary. If d is the thickness of the region of B, the angle θ through which the particle deviates from the initial direction on leaving the region is given by :-

(1) $\sin\theta = Bd\left(\frac{q}{2mV}\right)^{1/2}$

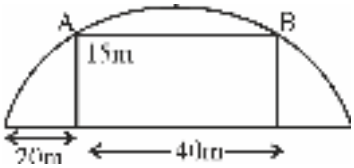
(2) $\cos\theta = Bd\left(\frac{q}{2mV}\right)^{1/2}$

(3) $\tan\theta = Bd\left(\frac{q}{2mV}\right)^{1/2}$

(4) $\cot\theta = Bd\left(\frac{q}{2mV}\right)^{1/2}$

21)

In the projectile motion shown in figure given time taken between points A and B is 2 sec. then maximum height will be :



(1) 40m

(2) 25m

(3) 20m

(4) 35m

22) Measure of two quantities along with the precision of respective measuring instrument is : A = 2.5 ms^{-1} ± 0.5 ms^{-1}, B = 0.10s ± 0.01 s
The value of AB will be

(1) (0.25 ± 0.08) m

(2) (0.25 ± 0.5) m

(3) (0.25 ± 0.05) m

(4) (0.25 ± 0.135) m

23)

The amplitude of electric and magnetic fields in a parallel beam of light of intensity 3.3 W/m^2 are,

respectively :-

(1) 50 N/C, 1.67×10^{-7} T

(2) 1.67×10^{-7} N/C, 50 T

(3) 18.19 N/C, 6 T

(4) 6 N/C, 18.19 T

24) If the radius of the earth reduces by 4% and the density remains same then escape velocity will :-

(1) reduce by 2%

(2) increase by 2%

(3) reduce by 4%

(4) increase by 4%

25) A point source of light is kept at a depth of h in water of refractive index 4/3. The radius of the circle at the surface of water through which light emits is :-

(1) $\frac{3}{\sqrt{7}}h$

(2) $\frac{\sqrt{7}}{3}h$

(3) $\frac{\sqrt{3}}{7}h$

(4) $\frac{7}{\sqrt{3}}h$

26) One mole of an ideal monoatomic gas at temperature T^0 expands slowly according to the law $\frac{P}{V}$ = constant. If the final temperature is $2T_0$, heat supplied to the gas is :-

(1) $2RT_0$

(2) RT_0

(3) $\frac{3}{2}RT_0$

(4) $\frac{1}{2}RT_0$

27) The speed of light (c) , gravitational constant (G) and Planck's constant (h) are taken as the fundamental units in a system. The dimension of time in this new system should be :

(1) $G^{1/2}h^{1/2}c^{-5/2}$

(2) $G^{-1/2}h^{1/2}c^{1/2}$

(3) $G^{1/2}h^{1/2}c^{-3/2}$

(4) $G^{1/2}h^{1/2}c^{1/2}$

28) What is the torque of force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ acting at a point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about the origin?

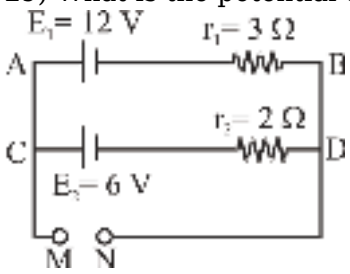
(1) $6\hat{i}-6\hat{j}+12\hat{k}$

(2) $-6\hat{i}+6\hat{j}-12\hat{k}$

(3) $17\hat{i}-6\hat{j}-13\hat{k}$

(4) $-17\hat{i}+6\hat{j}+13\hat{k}$

29) What is the potential difference between the points M and N for the circuit shown in figure



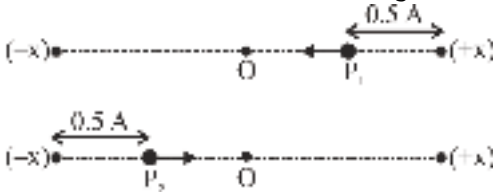
(1) 2.4

(2) 8.4 V

(3) 5.2 V

(4) 7 V

30) Two bodies performing S.H.M. have same amplitude and frequency. Their phases at a certain instant are as shown in the figure. The phase difference between them is :-



(1) $\frac{11}{6}\pi$

(2) π

(3) π/3

(4) $\frac{3}{5}\pi$

31) Diameter of a copper wire is measured using a screw gauge in which 4 full turns advances screw by 2mm on main scale and total number of divisions on circular scale is 100. Following observations are made : (a) When nothing is placed and screw is completely closed : Main scale reading : 0th division visible Circular scale reading : 6th division (b) When wire is placed to measure diameter : Main scale reading : 4 divisions Circular scale reading : 23rd division. The correct value of diameter is :

(1) 2.085 mm

(2) 2.145 mm

(3) 4.085 mm

(4) 4.018 mm

32) In an experiment to find the resistance of galvanometer by half deflection method, a 5V battery and a high resistance of 4.9 kΩ are connected with galvanometer. In absence of any shunt resistance

galvanometer reads 20 divisions when current flows in circuit. To reduce the deflection by half the value of shunt resistance used is 98 Ω. The figure of merit of galvanometer is given as :

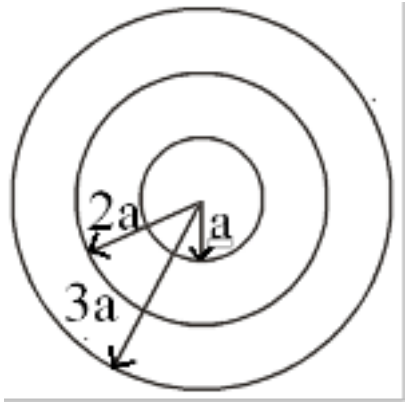
(1) 5 μA / division

(2) 20 μA / division

(3) 25 μA / division

(4) 50 μA / division

33) Three concentric metallic shell of radius a, 2a, 3a has charges Q, –2Q and 3Q respectively. When middle shell is earthed, then how much charge will flow from shell to earth :



(1) $-\frac{3Q}{2}$

(2) Q

(3) –Q

(4) zero

34) In an experiment kinetic energy K_{max} of photo electrons is 1.8 eV. If frequency of light is increased by 25% then K_{max} is doubled. Value of work function of metal is :-

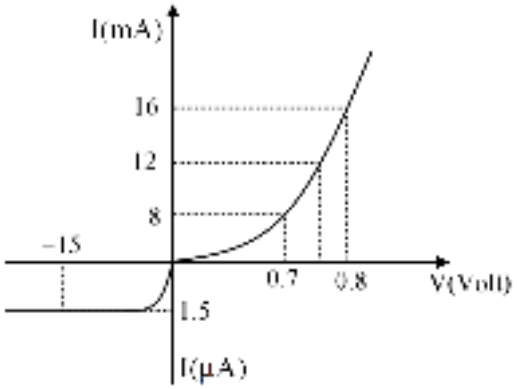
(1) 1.8 eV

(2) 3.6 eV

(3) 5.4 eV

(4) 9.0 eV

35) For a P-N junction diode its V–I characteristics are as shown in figure. The ratio of reverse to forward bias resistance is :



(1) 4×10^5

(2) 8×10^5

(3) 10^6

(4) 10^{-6}

36) The area of cross-section, length and density of a piece of a metal of atomic weight 60 g are 10^{-6} m^2 , 1.0 m and 5×10^3 kg/m^3 respectively, every atom contributes one free electron. (Given Avogadro number = 6×10^{23} / mol). Find the drift velocity of electrons in the metal when the current of 16 A passes through it.

(1) 2 mm/s

(2) 2 cm/s

(3) 20 cm/s

(4) 2 m/s

37) A wire of length l meter, made of a material of specific gravity 8 is floating horizontally on the surface of water. If it is not wet by water, the maximum diameter of the wire (in millimeters) upto which it can continue to float is: (Take surface tension of water T = 22.5 $\pi \times 10^{-3}$ N/m, g = 10 m/s^2) :-

(1) 1.5

(2) 1.1

(3) 0.75

(4) 0.55

38) Two liquids A and B have contact angles θ_A and θ_B in a glass capillary tube. If a factor k = $\cos\theta_A/\cos\theta_B$, then choose the correct statement.

(1) If k is positive, then liquid A has plane while liquid B has concave meniscus.

(2) If k = 0, then liquid A has concave meniscus and liquid B has convex meniscus.

(3) If k is negative, then liquid A and liquid B have concave meniscus.

(4) If k is negative, then liquid A has concave meniscus and liquid B has convex meniscus.

39) For a certain organ pipe three successive resonant frequencies are observed at 425, 595 and 765 Hz respectively. Taking the speed of sound in air to be 340 m/sec.
(i) whether the pipe is closed end or open end.
(ii) determine the length of pipe.

(1) closed end, 1 m

(2) open end, 1 m

(3) closed end, 2m

(4) open end, 2 m

40) A ball collides head on with another stationary ball of double mass with a velocity of 1.5 m/s. If the coefficient of restitution is 0.6 then after the collision their velocities will be :

(1) –0.1 m/s and 0.8 m/s

(2) 0.8 m/s and 0.1 m/s

(3) 0.1 m/s and 8 m/s

(4) 8 m/s and 0.1 m/s

41) A solenoid of length 50 cm, having 100 turns carries a current of 2.5 A. The magnetic field at one end of the solenoid is :-

(1) 3.14×10^{-4} T

(2) 6.28×10^{-4} T

(3) 1.57×10^{-4} T

(4) 9.42×10^{-4} T

42) In given fusion reaction :

$_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + n + 3.2$ MeV

The energy released in the fusion of 2 kg deuterons is approx :-

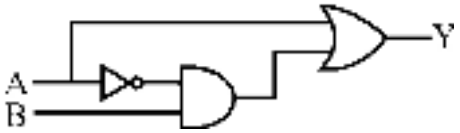
(1) 10^{30} eV

(2) 5×10^{23} MeV

(3) 10^{27} Mev

(4) 10^{33} Mev

43) In the following circuit, the output Y for all possible inputs A and B is expressed by the truth table :-



(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

44) A small loop of wire of area A = 0.01 m^2, N = 40 turns and resistance R = 20Ω is initially kept in a uniform magnetic field B in such a way that the field is normal to the plane of the loop. When it is pulled out of the magnetic field, a total charge of Q = 2×10^{-5} C flows through the coil. The magnitude of the field B is

(1) 1×10^{-3} T

(2) 4×10^{-3} T

(3) zero

(4) Unobtainable, as the data is insufficient

45) The bulk modulus of copper is 138×10^9 Pa. The pressure in an explosion chamber is 345×10^6 Pa. Then the percentage change in the volume of a piece of copper placed in this chamber will be :-

(1) 0.5%

(2) 0.25%

(3) 0.75%

(4) 2%

CHEMISTRY



1) The probability density plots of 1s and 2s orbitals are given in figure: The density of dots in a region represents the probability density of finding electrons in the region. On the basis of above diagram which of the following statements is/are correct?
(i) 1s and 2s orbitals are spherical in shape.
(ii) The probability of finding the electron is minimum near the nucleus.
(iii) The probability of finding the electron at a given distance is equal in all directions.
(iv) The probability density of electrons for 2s-orbital decreases uniformly as distance from the nucleus increases.

(1) (i) & (ii) only

(2) (ii) & (iii) only

(3) (i) & (iii) only

(4) only (iv)

2) Equal moles of H_2 and I_2 are heated in a closed vessel till equilibrium is obtained. If 80% of the hydrogen is converted to HI, the K_c at this temperature is ($H_2 + I_2 \rightleftharpoons 2HI$)

(1) 64

(2) 16

(3) 0.25

(4) 14

3) What is the molar solubility of Ag_2CO_3 ($K_{sp} = 4\times10^{-13}$) in 0.1 M Na_2CO_3 solution :-

(1) 10^{-6}

(2) 10^{-7}

(3) 2×10^{-6}

(4) 2×10^{-7}

4) How many electrons are present in 2×10^{-3} mole of ${}^{18}_{8}O^{2-}$?

(1) 1.2×10^{21}

(2) 9.6×10^{21}

(3) 1.2×10^{22}

(4) 1.9×10^{22}

5) $KMnO_4$ reacts with ferrous ammonium sulphate according to the equation :
$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$, here 10 mL of 0.1 M $KMnO_4$ is equivalent to :

(1) 20 mL of 0.1 M $FeSO_4$

(2) 30 mL of 0.1 M $FeSO_4$

(3) 40 mL of 0.1 M $FeSO_4$

(4) 50 mL of 0.1 M $FeSO_4$

6)

In the reaction
$Cl_2 + OH^- \rightarrow Cl^- + ClO_3^- + H_2O$
Chlorine is :-

(1) Oxidised

(2) Reduced

(3) Oxidised as well as reduced

(4) Neither oxidised nor reduced

7) The standard enthalpies of formation of $CO_{2(g)}$, $H_2O_{(\ell)}$ and glucose$_{(s)}$ at 25°C are – 400 KJ/mole, –300 KJ/mole and –1300 KJ/mol respectively. The standard enthalpy of combustion of glucose at 25°C (per mole).

(1) –2900 KJ/mole

(2) +2900 KJ/mole

(3) –1100 KJ/mole

(4) +1100 KJ/mole

8)

In which of the following processes entropy increases ?
A. A liquid evaporates to vapour
B. Temperature of a crystalline solid lowered from 130 K to 0K.
C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$
D. $Cl_{2(g)} \rightarrow 2\ Cl_{(g)}$
Choose the **correct** answer from the options given below :

(1) A and C

(2) A, B and D

(3) A, C and D

(4) C and D

9) $A_2 + B_2 \rightarrow 2AB$; R.O.R = $k[A_2]^a [B_2]^b$

Initial[A_2]	**Initial[B_2]**	**R.O.R(r) Ms^{-1}**
0.2	0.2	0.04
0.1	0.4	0.04
0.2	0.4	0.08

Order of reaction with respect to A_2 and B_2 are respectively :

(1) a = 1, b = 1

(2) a = 2, b = 0

(3) a = 2, b = 1

(4) None of these

10) Rate of a reaction can be expressed as :
$r = PZ_{AB}\, e^{-Ea/RT}$
For above expression, match the following:

(a)	P	(i)	Activation energy
(b)	Z_{AB}	(ii)	Fraction of activated molecules
(c)	$e^{-Ea/RT}$	(iii)	Probability or steric factor
(d)	E_a	(iv)	Collision frequency

Choose the correct answer from following options:

(1) (a)-(i) , (b)-(ii) , (c)-(iii) , (d)-(iv)

(2) (a)-(iii) , (b)-(ii) , (c)-(iv) , (d)-(i)

(3) (a)-(iii) , (b)-(iv) , (c)-(ii) , (d)-(i)

(4) (a)-(iv) , (b)-(i) , (c)-(ii) , (d)-(iii)

11) Given below are two statements:
Statement-I : The electrolysis of molten sodium hydride liberates H_2 gas at the anode.
Statement-II : In electrolytic refining of an impure metal, the pure metal is obtained at anode.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and II are incorrect.

(2) Statement I is incorrect but statement II is correct.

(3) Statement I is correct but statement II is incorrect

(4) Both statement I and II are correct.

12) Given that $E^{o}_{Zn^{+2}|Zn}$ = –0.763 volt and
$E^{o}_{Fe^{+2}|Fe}$ = –0.44 volt, then the E.M.F. of the cell

$Zn_{(s)}\left|Zn^{+2}\right._{(0.001\ M)}\left\|Fe^{+2}(0.005M)\right|Fe_{(s)}$ is :-

(1) Equal to 0.323 volt

(2) Less than 0.323 volt

(3) Greater than 0.323 volt

(4) None of these

13) Osmotic pressure of a solution is 0.0821 atm at a temperature of 300 K. The concentration of solution in mol/L will be :-

(1) 0.33

(2) 0.666

(3) 0.3×10^{-2}

(4) 3

14)

Freezing point of an aqueous solution is – 0.325°C. Calculate boiling point of solution ?
[K_f = 1.86 K.Kgmol^{-1}, K_b = 0.52 K.Kgmol^{-1}]

(1) 100.09°C

(2) 101.5°C

(3) 100°C

(4) 103°C

15) Around 57 kJ mol^{-1} heat is released during which of the following reaction-

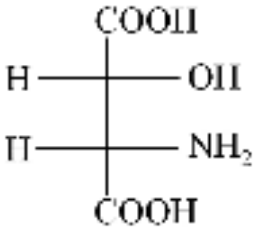
(1) $H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(\ell)}$

(2) $CH_3COOH_{(aq)} + NH_4OH_{(aq)} \rightarrow CH_3COONH_{4(aq)} + H_2O_{(\ell)}$

(3) $HCl_{(aq)} + NaOH_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(\ell)}$

(4) $HCN_{(aq)} + NaOH_{(aq)} \rightarrow NaCN_{(aq)} + H_2O_{(\ell)}$

16) Correct IUPAC name –



(1) (2S, 3R) – 2 – amino – 3 – hydroxy butane – 1, 4 – dioic acid

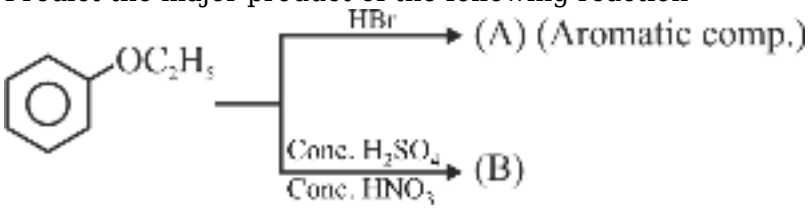
(2) (2R, 3S) – 2 – amino – 3 – hydroxy butane – 1, 4 – dioic acid

(3) (2R, 3R) – 2 – hydroxy – 3 – amino butane – 1, 4 – dioic acid

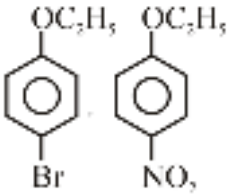
(4) (2R, 3S) – 2 – hydroxy – 3 – amino butane – 1, 4 – dioic acid

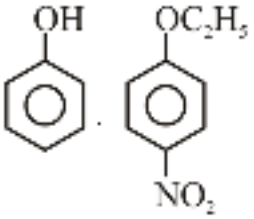
17)

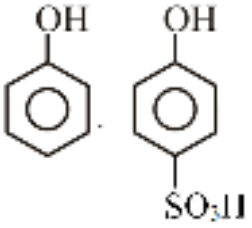
Predict the major product of the following reaction

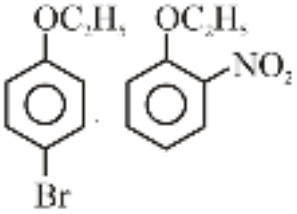


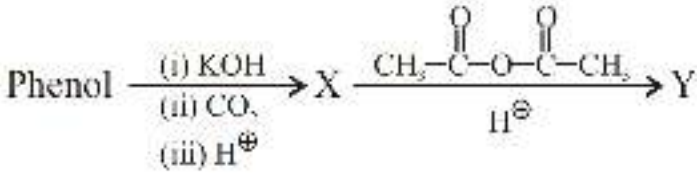
Choose the option with appropriate products A & B respectively are :

(1) 

(2) 

(3) 

(4) 

18) 

Y is :

(1) Acetyl salicylic acid

(2) Methyl acetyl salicylate

(3) Phenyl acetate

(4) Phenyl salicylate

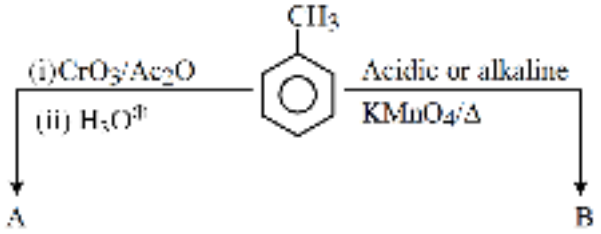
19) C_6H_{10} number of structural isomer of (alkyne only) :-

(1) 4

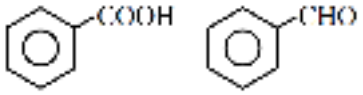
(2) 5

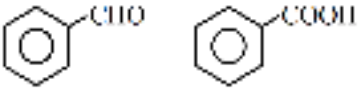
(3) 6

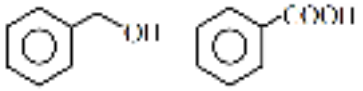
(4) 7

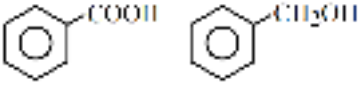
20) 

Structure of A and B respectively are :

(1) 

(2) 

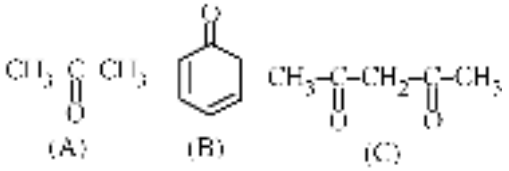
(3) 

(4) 

21) Which of the following statement is incorrect :

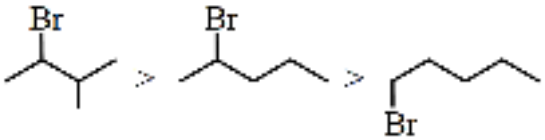
(1) Aliphatic aldehyde gives silver mirror with tollen's reagent.
(2) Ketone gives yellow – orange ppt with 2,4 – DNP
(3) Terminal alkyne gives white salt with tollen's reagent
(4) Aromatic aldehyde form red-brown ppt with fehling solution

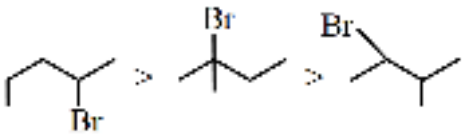
22) Correct order of percentage enol content is-

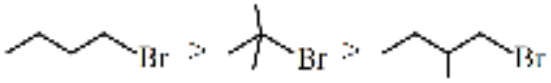


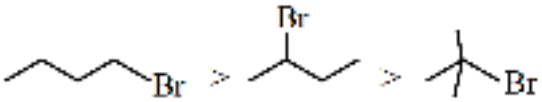
(1) B>C>A
(2) A>B>C
(3) C>B>A
(4) B>C≈A

23) Which of the following is correct decreasing order for S_N2 reaction ?

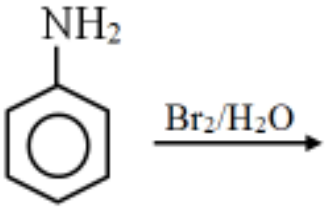
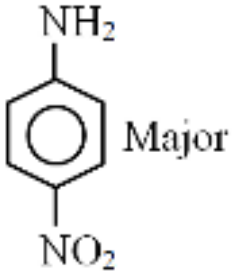
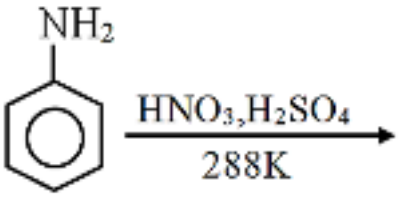
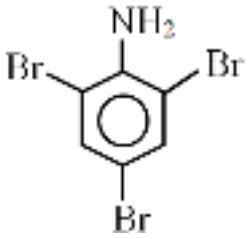
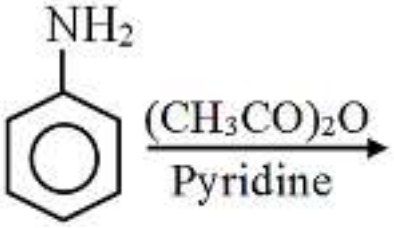
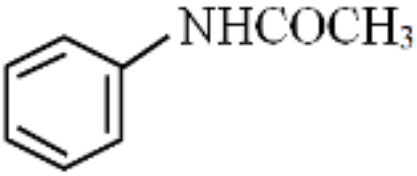
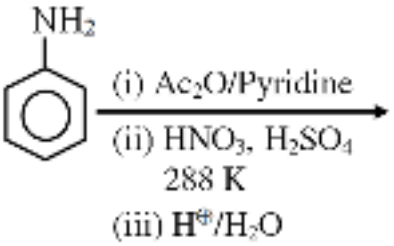
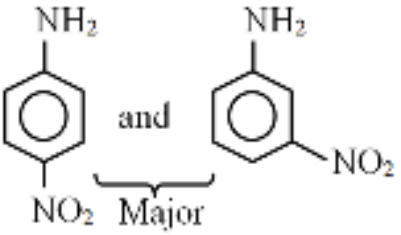
(1) 

(2) 

(3) 

(4) 

24)

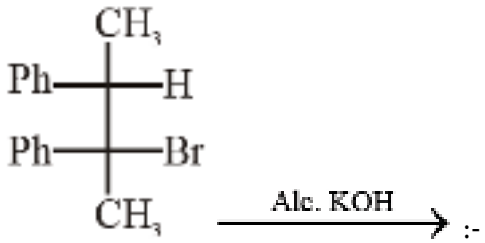
	Column-I (Reactant)		Column-II (Product)
A		P	
B		Q	
C		R	
D		S	

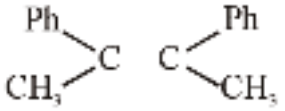
(1) A - P, B - Q,C - R, D - S

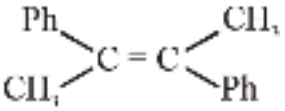
(2) A - Q, B - P,C - R, D - S

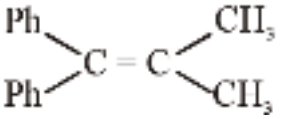
(3) A - S, B - P,C - Q, D - R

(4) A - Q, B - S, C - R, D - P

25)  :-

(1) 

(2) 

(3) 

(4) 

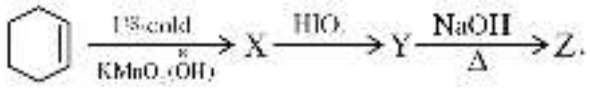
26)  Find hybridisation at C_1, C_4, C_7 respectively?

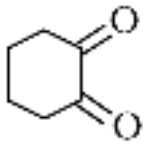
(1) sp^2, sp^2, sp

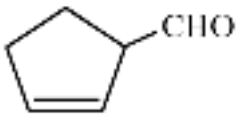
(2) sp^2, sp, sp^2

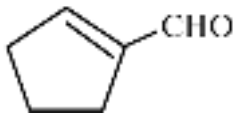
(3) sp, sp^2, sp

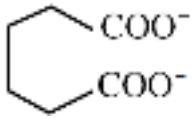
(4) sp, sp^2, sp^2

27)  Z is :

(1) 

(2) 

(3) 

(4) 

28) What are hydrolysis products of lactose

(1) β-D galactose and β-D glucose

(2) α-D glucose and α-D glucose

(3) α-D glucose and β-D fructose

(4) β-D glucose and β-D galactose

29) **Statement I :** In Duma's method when an organic compound is heated with cupric oxide, N is converted to N_2 gas
Statement II : Cupric oxide oxidized carbon & hydrogen to CO_2 and H_2O

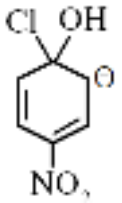
(1) Both Statement I and II are correct

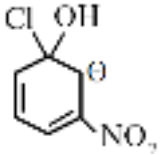
(2) Both Statement I and II are incorrect

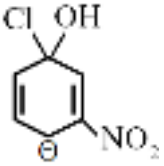
(3) Statement I is correct but statement II is incorrect.

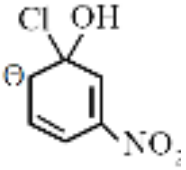
(4) Statement I is incorrect but statement II is correct.

30) Which carbanion is highly stable ?

(1) 

(2) 

(3) 

(4) 

31) Match the column :-

Element		Electron gain Enthalpy (ΔHeg) kJ/mole	
(a)	He	(i)	+ 96
(b)	Ne	(ii)	+ 48
(c)	Ar	(iii)	– 122
(d)	C	(iv)	– 328
(e)	F	(v)	+ 116

(1) (a) – (ii), (b) – (iii), (c) – (v), (d) –(iv), e – (i)

(2) (a) – (ii), (b) – (i), (c) – (v), (d) –(iv), e – (iii)

(3) (a) – (ii), (b) – (v), (c) – (i), (d) –(iii), e – (iv)

(4) (a) – (i), (b) – (ii), (c) – (v), (d) –(iii), e – (iv)

32) Match the following.

Column-I (size)		Column-II (Reason)	
(i)	$Na^{+} > Mg^{+2}$	(a)	Lanthanoide Contraction
(ii)	Al ≥ Ga	(b)	Isotope
(iii)	Ru ≈ Os	(c)	Transition Contraction
(iv)	$Cl^{35} \approx Cl^{37}$	(d)	Z_{eff}

(1) (i)-d ; (ii)-c ; (iii)-b ; (iv)-a

(2) (i)-d ; (ii)-b ; (iii)-a ; (iv)-c

(3) (i)-d ; (ii)-c ; (iii)-a ; (iv)-b

(4) (i)-c ; (ii)-d ; (iii)-a ; (iv)-b

33) Which of the following molecule exist in vapour phase at high temperature.

(1) Be_2

(2) C_2

(3) O_2

(4) N_2

34) Which of the following is incorrect -

(1) Viscosity order $H_2O > CH_3OH > CH_3 – O – CH_3$

(2) Boiling point order $H_2O > HF > NH_3$

(3) Molecular weight of acetic acid is double of its molecular formula in non polar solvent is due to dimer formation by H-bonding.

(4) In O-Salicylaldehyde inter molecular H-bonding is present.

35) **Statement I :** The larger cation and the larger size of anion, the greater the covalent character of an ionic bond.
Statement II : The Vander waal's radius represents the overall size of the atom which includes its valence shell in a non-bonded situation.

(1) Statement I and II are correct

(2) Statement I and II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement II is correct but Statement I is incorrect

36)

Which of the following is correct order of dipole moment (μ)?

(1) $HCl < H_2S < HBr < CHCl_3$

(2) $CHCl_3 < H_2S < HBr < HCl$

(3) $H_2S < CHCl_3 < HBr < HCl$

(4) $HBr < H_2S < CHCl_3 < HCl$

37) A complex $K_n [CoF_6]$ has magnetic moment 4.90 B.M. What will be the oxidation state of Co and the value of n?

(1) Co (II); n = 3

(2) Co (III); n = 3

(3) Co (III); n = 2

(4) Co (III); n = 4

38) Select incorrect match for $[M(H_2O)_6]^{2+}$ complex.

	Metal Ions	**Electronic configuration**	**CFSE**
(1)	Mn^{+2}	$t_{2g}^3\ eg^2$	0
(2)	Fe^{+2}	$t_{2g}^4\ eg^2$	$-0.4\Delta_0$
(3)	Co^{+2}	$t_{2g}^5\ eg^2$	$-0.8\Delta_0$
(4)	Ni^{2+}	$t_{2g}^6\ eg^2$	$-1.6\Delta_0$

(1) 1

(2) 2

(3) 3

(4) 4

39) Choose the correct statements.
(i) Primary valency is normally non ionisable and satisfied by negative ions
(ii) Secondary valency is equal to C.N.
(iii) When $NiCl_2.6H_2O$ (one mole) Reacts with excess $AgNO_3$ then 2 mole of AgCl precipitated.
(iv) Secondary Valency is ionisable and satisfied by neutral molecules or negative ions.

(1) (i), (iv)

(2) (i), (ii), (iv)

(3) (ii), (iii)

(4) (iii) (iv) (i)

40) **Assertion (A) :** Copper (I) compounds are unstable in aqueous solution and undergo disproportionation.
Reason (R) : Cu^{2+}(aq.) is more stable than Cu^{+}(aq.) due to much more negative $\Delta_{hy}H$ of Cu^{2+} which more than compensates for second ionisation enthalpy of Cu.
Which of the following option is correct-

(1) (A) is correct but (R) is incorrect.

(2) (A) is incorrect but (R) is correct.

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

41) The complex formation tendency of transition metal ions is too high because :-

(1) Comparatively smaller ionic size

(2) High ionic charges.

(3) Availability of vacant d-orbital for coordinate bond formation

(4) All of these

42) Match the following Column -

Column-I (Substance)		**Column-II (Application)**	
(A)	Helium	(I)	Bulb designed for special purpose
(B)	Argon	(II)	Advertising sign
(C)	Neon	(III)	Inert Atmosphere
(D)	Xenon	(IV)	Cryogenics agent

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-III, D-IV

43) Which statement about white phosphorous is correct ?

(1) White phosphorous contain 60° bond angle

(2) It dissolve in boiling NaOH solution in an inert atmosphere giving PH_3

(3) It readily catches fire to give dense white fumes at P_4O_{10}

(4) All of these

44) Which of the following statement regarding mohr salts is not true ?

(1) Oxidation state of Fe in mohr salt is +2.

(2) Formula of Mohr salt is $(NH_4)_2SO_4.FeSO_4.6H_2O$.

(3) Dilute sulfuric acid is used to prepare mohr salt.

(4) The appearance of mohr's salt is white in colour.

45) Mark the incorrect statement out of the following :

(1) Group I basic radical precipitate as chlorides.

(2) Group IV basic radical precipitate as sulphides.

(3) Group V basic radical precipitate as carbonates.

(4) Group II basic radical precipitate as hydroxide.

BIOLOGY

1) Most names in nomenclature are derived from.

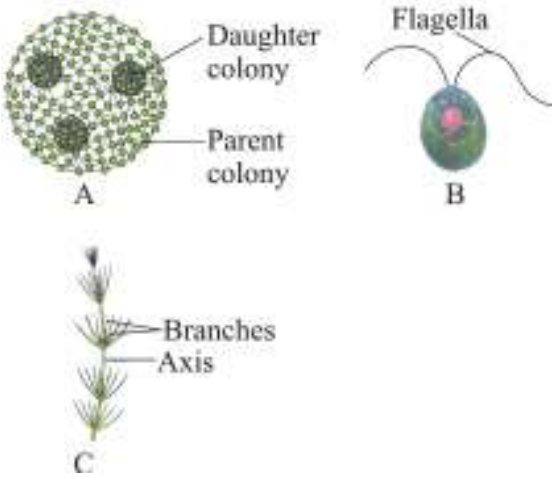
(1) English

(2) Hindi

(3) German

(4) Latin

2) Identify the given figures of algae and select the correct option.



(1) A–*Chlamydomonas*, B–*Chara*, C–*Volvox*

(2) A–*Volvox*, B–*Chlamydomonas*, C–*Chara*

(3) A–*Chara*, B–*Laminaria*, C–*Volvox*

(4) A–*Porphyra*, B–*Polysiphonia*, C–*Fucus*

3) What is Bentham and Hooker classification demerit ?

(1) Bentham and Hooker classification is natural classification

(2) Bentham and Hooker classification was mainly based on floral characters

(3) Bentham and Hooker classification is based on actual observation

(4) In Bentham and Hooker classification the phylogeny of plants is not considered

4)

Read the following statements about mycoplasma:-
(I) It is pleomorphic organism which lack cell wall
(II) It is the smallest living organism
(III) They can survive without oxygen.
(IV) They are pathogenic in animals and plants.

Choose the **correct** statements :-

(1) I, II & III only

(2) II, III and IV only

(3) I, II & IV only

(4) I, II, III and IV all

5)

Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.
A. Multicellular heterotrophs with cell wall made of chitin.
B. Heterotrophs with tissue/organ/organ system level of body organization.
C. Prokaryotes with cell wall made of polysaccharides and amino acids.
D. Eukaryotic autotrophs with tissue/organ level of body organization.
E. Eukaryotes with cellular body organization.
Choose the **correct** answer from the options given below :

(1) A, C, E, B, D

(2) C, E, A, D, B

(3) A, C, E, D, B

(4) C, E, A, B, D

6) Read statements A to D :
(A) Golden Algae are classified under chrysophytes.
(B) *Gonyaulax* & Diatoms are classified under Dinoflagellates
(C) In Diatoms the cell wall form thin overlapping shell, which fit together as in soap box.
(D) Most of the Dinoflagellates have two flagella.
How many statement are **wrong** ?

(1) one

(2) two

(3) three

(4) four

7) **Statement-I :** Roots coming out of the base of the stem is called fibrous root.
Statement-II : Fibrous types of roots are found in wheat.

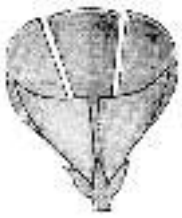
(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

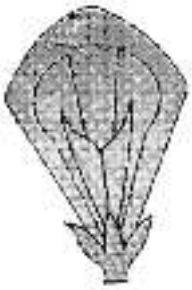
(4) Both Statement-I & Statement-II are incorrect.

8) Which of the following aestivation of petals is shown by a family having ten, diadelphous, dithecous stamens and legume fruit :-

(1) 

(2) 

(3) 

(4) 

9) Match the three columns correctly and find the correct option.

	Column-I		**Column-II**		**Column-III**
(A)	Parallel venation	(i)	Mustard	(P)	Liliaceae
(B)	*Calotropis*	(ii)	Acropetal succession	(Q)	Brassicaceae
(C)	Racemose	(iii)	Monocots	(R)	Opposite leaves
(D)	False septum	(iv)	Valvate	(S)	Indefinite growth of main axis

(1) A-ii-P, B-i-Q, C-i-Q, D-ii-P

(2) A-i-P, B-ii-P, C-ii-R, D-iii-S

(3) A-iii-Q, B-ii-R, C-i-S, D-iv-R

(4) A-iii-P, B-iv-R, C-ii-S, D-i-Q

10) Vascular bundles in monocotyledons stem are considered closed because :-

(1) There are no vessels with perforations

(2) Cambium is present

(3) Cambium is absent

(4) Xylem surrounded all around by phloem

11) The transverse section of plant material shows the following anatomical features under microscope: Palisade parenchyma towards adaxial surface and spongy parenchyma towards abaxial

surface. Conjoint, collateral and closed vascular bundle and xylem is endarch. Two distinct patches of parenchyma are present above and below, each of the large vascular bundle.
Plant material should be :-

(1) Isobilateral leaf
(2) Dorsiventral leaf
(3) Monocotyledonae stem
(4) Dicotyledonae stem

12) Select the correct match in the following pairs :-

(1) Peroxisome – oxidation of glucose
(2) Lysosome – synthesis of lipids
(3) Smooth ER – synthesis of protein
(4) Rough ER – Synthesis of protein



13) Which of the following pair is correct for given diagram ?

	Diagram is	**Involve in**
(1)	Nuclear membrane	DNA Synthesis
(2)	E.R.	Steroid Synthesis
(3)	Golgi complex	Glycoprotein formation
(4)	E.R.	Protein synthesis

(1) 1
(2) 2
(3) 3
(4) 4

14) The cell organelle which is responsible to create luminal and extra-luminal compartment in eukaryotic cell is :-

(1) Endoplasmic reticulum
(2) Golgi complex
(3) Nucleus
(4) Vacuole

15) Which sub-stage of prophase-I is characterized by appearance of recombination nodule ?

(1) Zygotene

(2) Pachytene

(3) Diplotene

(4) Leptotene

16) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The individual chromosomes can no longer be seen and chromatin material tends to collect in a mass at two poles in Telophase.
Reason (R) : The chromosomes that have reached their respective poles decondense and lose their individuality.
In the light of the above statements. Choose the correct answer from the options given below :

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct.

(3) (A) and (R) both are incorrect.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

17) Which one is correct?

(1) Inorganic catalysts work efficiently at high temperature and high pressure while enzymes get damaged at high temperature (above 40°C)

(2) Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80 - 90°C)

(3) Ribozymes are nucleic acids behaving like enzyme

(4) All are correct

18) The scientist who proved that bacteria use H_2S and CO_2 to synthesize carbohydrate is :

(1) Van Niel

(2) Ruben

(3) Jean Senebier

(4) Julius Robert Mayer

19) Which of the following hastens fruit ripening in tomatoes and apples :-

(1) Ethephone

(2) Kinetin

(3) Abscisic acid

(4) 2, 4D

20) **Assertion :** The maximum concentration of alcohol in beverages that are naturally fermented is about 13 percent.
Reason : Fermenting organism like yeast poison themselves to death when the concentration of alcohol reaches about 13 percent.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

21) **Statement - I :-** In one turn of TCA cycle, only two molecules of ATP are formed.
Statement - II :- Complete oxidation of acetyl - Co - A occurs during TCA cycle.

(1) Both statement - I and Statement - II are correct.

(2) Both statement - I and Statement - II are incorrect.

(3) Statement - I is correct but Statement - II is incorrect.

(4) Statement - I is incorrect but Statement - II is correct.

22) To initiate flowering and synchronizing the fruit set in pineapple which can be used :-

(1) ABA

(2) Ethylene

(3) Cytokinin

(4) Gibberellin

23) Geitonogamy is :

(1) Functionally cross pollination involving pollinating agent as well as genetically it is cross pollination since the pollen grain come from another plant

(2) Functionally self pollination it involves no pollinating agent and genetically it is similar to cross pollination since the pollen grain come from another plant

(3) Functionally cross pollination involving pollinating agent and genetically it is similar to autogamy since the pollen grain come from the same plant

(4) Functionally self pollination, it requires no pollinating agent and genetically it is similar to autogamy since the pollen grain come from the same flower

24) Monoecious plant with unisexual flowers prevents :-

(1) Autogamy

(2) Geitonogamy

(3) Both autogamy and geitonogamy

(4) Pollination is not possible

25) Which of the following statement is not correct :-

(1) In autogamy, pollination is achieved within same flower.

(2) In embryosac, three cells are present toward the chalazal end are called egg apparatus

(3) Polar nuclei are situated below the egg apparatus.

(4) Megaspore develops into the female gametophyte.

26) How many parts in list given below are related to structure of ovule :-
Stomium, Anther, Germpore, Chalaza, Tapetum, Nucellus, Micropyle, Raphae, Embryosac

(1) 7

(2) 6

(3) 5

(4) 4

27) A test cross of F_1 flies +a/+b produced the following offspring :-
++/ab = 9 ab/ab = 9
+b/ab = 41 a+/ab = 41
What will be distance between linked gene :–

(1) 82 cM

(2) 18 cM (cis)

(3) 20 cM

(4) 18 cM (trans)

28) Which statement is not true for *Drosophila melanogaster* ?

(1) They complete their life cycle in about two weeks

(2) Single mating produce large number of progeny flies

(3) It has few hereditary variations that can only be seen with high power microscope

(4) It has clear differentiation of the sex

29) F_1 generation when resembles :-
A - either of the two parents
B - in between of the two parents
C - with both of the parents
Choose correct match :-

	A	**B**	**C**
(1)	Dominance	Incomplete dominance	Co-dominance
(2)	Dominance	Co-dominance	Incomplete dominance
(3)	Incomplete dominance	Co-dominance	Dominance
(4)	Co-dominance	Dominance	Incomplete dominance

(1) 1

(2) 2

(3) 3

(4) 4

30) Which is responsible for cytoplasmic inheritance :-

(1) Mitochondria

(2) Chloroplast

(3) Mitochondria and chloroplast

(4) Mitochondria, chloroplast and nucleus

31) Morgan carried out several dihybrid crosses in Drosophila and discovered Linkage. He observed that :

(1) Genes which were tightly linked showed higher recombination.

(2) Gene of white eyes and yellow body in Drosophila were tightly linked and showed 1.3% recombination.

(3) Genes which were loosely linked showed higher linkage.

(4) Genes of white eyes and miniature wings in drosophila showed 37.2% linkage.

32) Which set of RNA are involved in protein synthesis ?

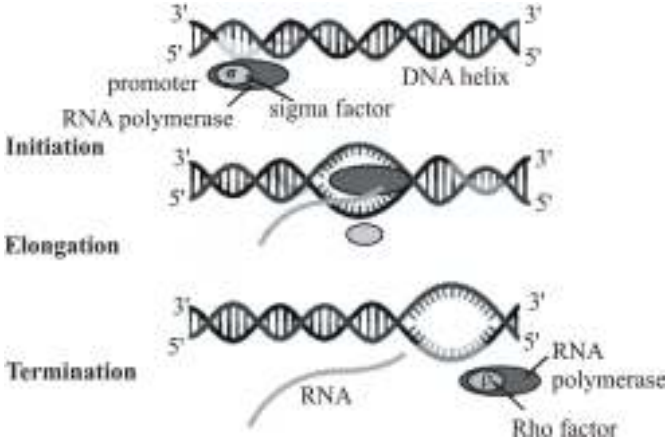
(1) tRNA, mRNA, rRNA

(2) tRNA, mRNA, hnRNA

(3) hnRNA, mRNA, rRNA

(4) hnRNA, tRNA, rRNA

33) Given below diagram represents which of the following process ?



(1) Replication

(2) Transcription in Eukaryotes

(3) Transcription in Prokaryotes

(4) Translation

34) In which respect RNA synthesized on DNA template in nucleus differ from template :

(1) in segment transcribed from exon

(2) in segment transcribed from intron

(3) in presence of Uracil

(4) in presence of Thymine

35) Which of the following match is correct.

	Organism		No. of nucleotide
(i)	$\phi \times 174$ Bacteriophage	(A)	4.6×10^6 bp
(ii)	Lambda bacteriophage	(B)	3.3×10^9 bp
(iii)	*E.coli*	(C)	5386 N bases
(iv)	Human	(D)	48502 bp

(1) (i) = D, (ii) = C, (iii) = B, (iv) = A

(2) (i) = B, (ii) = D, (iii) = A, (iv) = C

(3) (i) = A, (ii) = D, (iii) = C, (iv) = B

(4) (i) = C, (ii) = D, (iii) = A, (iv) = B

36) Match **List-I** with **List-II.**

	List-I		List-II
A.	Alfred Hershey and Martha Chase	I.	Streptococcus pneumoniae
B.	Euchromatin	II.	Densely packed and dark-stained
C.	Frederick Griffith	III.	Loosely packed and light-stained
D.	Heterochromatin	IV.	DNA as genetic material confirmation

Choose the **correct** answer from the options given below :

(1) A–II, B–IV, C–I, D–III

(2) A–IV, B–II, C–I, D–III

(3) A–IV, B–III, C–I, D–II

(4) A–III, B–II, C–IV, D–I

37) Match the following w.r.t HGP

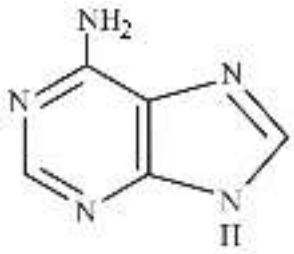
Column-I		Column-II	
(a)	ESTs	(I)	Coding and non-coding DNA sequence
(b)	Sequence annotation	(II)	Computational techniques for genome sequencing
(c)	Satellite DNA	(III)	Gene which is expressed as RNA
(d)	Bio-informatics	(IV)	Non-coding DNA

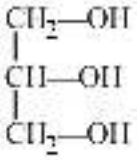
(1) a-(III), b-(I), c-(IV), d-(II)

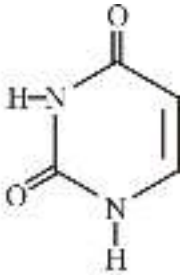
(2) a-(IV), b-(I), c-(III), d-(II)

(3) a-(IV), b-(III), c-(I), d-(II)

(4) a-(III), b-(IV), c-(II), d-(I)







38) (A) (B) (C)
Which of the above may be present in both RNA & DNA ?

(1) A
(2) B
(3) C
(4) Both A & B

39) How many examples given below are not the example of detrimental relations.
Epiphyte on mango, Lichen, Mycorrhizae, Cuscuta on Hedge plant, Barnacle on whale, Ticks on dog :-

(1) Four
(2) Six
(3) Two
(4) Five

40) Match the column and choose the correct option

	Column-I		**Column-II**
(a)	Birds	(i)	Breed only once in their life time
(b)	Oysters	(ii)	Breed many time during life time
(c)	Bamboo	(iii)	Produce a large number of small size offspring
(d)	Mammals	(iv)	Produce a small number of large size offspring

(1) a-i, b-ii, c-iii, d-iv
(2) a-iv, b-iii, c-i, d-ii
(3) a-iii, b- iv, c-ii, d-i
(4) a-i, b-ii, c-iv, d-iii

41) How many statements are correct ?
(a) Net primary productivity is available biomass for consumption to heterotrops
(b) Secondary productivity is defined as rate of formation of new organic matter by heterotrophs
(c) The annual net primary productivity of ocean is greater than land
(d) Warm and moist environment favours decomposition

(1) a, b, c, d
(2) a, b, c

(3) a, b, d

(4) a, c

42)

Read the following four statement (A-D) :-
(A) Geometric growth produces J-shaped population growth curve
(B) Logistic growth occurs when resources are limiting
(C) Integral form of equation for exponential growth is $N_t = N_0 e^{rt}$
(D) Limiting factors can influence the number of organisms in a population once if reaches its carrying capacity

How many of the above statements are right ?

(1) Four

(2) One

(3) Two

(4) Three

43) Death of organisms is the begining of :

(1) Grazing food chain

(2) Parasitic food chain

(3) Detritus food chain

(4) Both 1 and 2 given above

44) Which one of the following is an example of In-situ conservation?

(1) Sacred groves

(2) Botanical garden

(3) Wild life safari

(4) Cryopreservation

45) Sucrose is made up of :

(1) Two α-D glucose units

(2) α-D glucose & β-D fructose

(3) Two β-D fructose units

(4) α-D fructose & β-D fructose

46) Out of following which statement is incorrect for aves?

(1) The characteristics features of aves are the presence of feathers.

(2) The forelimbs are modified into wings.

(3) They are cold blooded animals.

(4) Air sacs connected to lungs supplement respiration.

47) Which of the following are correct statements for hemichordata ?

(A) They are bilaterally symmetrical triploblastic and coelomates.
(B) Circulation is of open type.
(C) Sexes are separate fertilisation is external and development is indirect.
(D) Possess a rudimentary structure in collar region.
(E) Excretory system is altogether absent.

(1) C and D
(2) B, C and D
(3) A, B, C and D
(4) B and E

48) Which of the following groups of animals shares similarity regarding maintenance of constant body temperature with mammals ?

(1) Reptiles
(2) Amphibians
(3) Aves
(4) Fishes

49) The symmetry shown by coelenterates and ctenophores is also exhibited by :-

(1) Annelida
(2) Echinoderms
(3) Molluscs
(4) Poriferans

50) Match column I with column-II can select the correct option :

Column-I		**Column-II**	
(a)	Petromyzon	(i)	Body with scales
(b)	Branchiostoma	(ii)	Ectoparasite on fishes
(c)	Doliolum	(iii)	Notochord persistent throughout life
(d)	Hippocompus	(iv)	Notochord present only in larval tail

(1) a-ii, b-iii, c-iv, d-i
(2) a-ii, b-iv, c-iii, d-i
(3) a-ii, b-iii, c-i, d-iv
(4) a-i, b-ii, c-iii, d-iv

51) The intercellular material of cartilage is :-

(1) Solid and pliable
(2) Solid and non pliable
(3) Solid and soft
(4) Solid and strong

52) In female cockroaches a pair of spermatheca is present in the _____.
Fill in the blank in the question given above.

(1) 6th to 7th abdominal segments.

(2) 6th thoracic segment

(3) 6th Abdominal segment

(4) 4th to 6th abdominal segments

53) **Statement-I :** The blood vascular system of cockroach is of open type including heart and well developed blood vassels.
Statement-II : In cockroach, blood from sinuses enter heart through ostia and is pumped anteriorly to sinus again.
Choose the correct answer from the options given below :

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I correct but statement-II is incorrect.

(3) Both statement-I and statement-II are correct.

(4) Statement-I is incorrect but statement-II is correct.

54) Which of the following statement correct regarding frog.

(1) Ventricle opens into a sac like conus arteriosus on the dorsal side of heart.

(2) Sinus venosus receive deoxy blood from right atrium.

(3) Bidder's canal is present in male frog kidney.

(4) Copulatory ped is present on first digit of hind limb in male frog.

55) On the ventral side of the heart of frog ventricle opens into : -

(1) Sinus venosus

(2) Pericardium

(3) Conus venosus

(4) Conus arteriosus

56) Least number of white blood corpuscles are :-

(1) Neutrophils

(2) Basophils

(3) Monocytes

(4) Eosinophils

57) Choose an agranulocyte which is phagocytic in nature ?

(1) Neutrophils

(2) Monocytes

(3) Both 1 & 2

(4) Eosinophils

58) **Assertion :** Blood group O^- individuals are universal donors.
Reason : Blood group O^- individual RBCs lack all three type of antigens (A, B & Rh)

(1) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

59) These are two statements given.
Statement-I :- The minimum volume of air accomodated in the lungs at the end of a forced expiration is called residual volume.
Statement-II :- The maximum volume of air a person can breath out after forceful inspiration is called vital capacity of lungs.
Choose the correct option :-

(1) Statement-I and statement-II both are false.

(2) Statement-I is true but statement-II is false.

(3) Statement-I and statement-II both are true.

(4) Statement-I is false but statement-II is true.

60) **Assertion :-** Meena a 50 year old women working in a stone cutting factor for over 25 years, started experiencing difficulty in breath and persistent dry cough. A CT scan revealed extensive fibrosis in lungs, doctor concluoled that her condition was due to continuous in inhalah of fine silica dust.
Reason :- In stone grinding, cutting industries so mush dust is produce that the defense mechanism of the body cannot fully cope with this situation long exposure can give rise to inflammation leading to fibrosis and thus causes serious lung damage.

(1) Assertion and reason both are true and reason is correct explanation of assertion.

(2) Assertion and reason both are true but reason is not correct explanation of assertion.

(3) Assertion and reason both are incorrect.

(4) Assertion is true but reason is false.

61) Total amount of carbondioxide transported by medium other than R.B.Cs

(1) 20-25%

(2) 70%

(3) 7%

(4) 77%

62) In the given substance how many correctly eliminate by sebaceous glands -
Sterols, Hydrocarbons, NaCl, Urea, Waxes, Lactic acid, Bilirubin

(1) 2

(2) 6

(3) 3

(4) 5

63) Cephalochordate's excretory organ is

(1) Protonephridia
(2) Nephridia
(3) Malpighian
(4) Green gland

64) The cerebral cortex is referred to as the ___(A)____ due to its ____(B) appearance. The neuron ____(C)____ are concentrated here giving the colour :

	(A)	(B)	(C)
(1)	White mater	White	Cell bodies
(2)	Grey mater	Grey	Axon
(3)	White mater	White	Axon
(4)	Grey mater	Grey	Cell bodies

(1) 1
(2) 2
(3) 3
(4) 4

65) The axonal membrane is _____ to negatively charged proteins present in the axoplasm.

(1) Selectively permeable
(2) permeable
(3) Semipermeable
(4) Impermeable

66) Read following terminology of column-I and column-II and Identify their correct match :-

	Column-I		**Column-II**
(A)	Unipolar neuron	(i)	Sympathetic neural system
(B)	Bipolar neuron	(ii)	One axon 2 two or more dendrile
(C)	Multipolar neuron	(iii)	Found in retina of eye
(D)	Autonomic neural system	(iv)	Cell body with one axon
		(v)	One axon and one dendrile
		(vi)	Found in Embryonic stage
		(vii)	Found in cerebral coalex
		(viii)	Parasympathetic neural system

(1) A-(iv), (vi), B-(ii), (vii), C-(iii), (v), D-(i), (viii)

(2) A-(iv), (vi), B-(iii), (v), C-(ii), (vii), D-(i), (viii)

(3) A-(vii), (viii), B-(v), (vi), C-(ii), (iv), D-(i), (ii)

(4) A-(ii), (vii), B-(iii), (v), C-(iv), (vi), D-(i), (viii)

67) Match the Column-I with Column-II and choose correct option :

	Column-I (Gland)		**Column-II (Function)**
A.	Pineal	1.	Regulates biological clock
B.	Hypothalamus	2.	Controls pituitary secretion
C.	Posterior pituitary	3.	Release oxytocin and ADH
D.	Anterior pituitary	4.	Secretes GH, TSH, FSH etc.

(1) A-1, B-3, C-4, D-2

(2) A-3, B-1, C-4, D-2

(3) A-1, B-2, C-3, D-4

(4) A-4, B-3, C-3, D-4

68) Oxytocin released from the posterior pituitary mainly affects.

(1) Blood sugar regulation

(2) Bone growth

(3) Uterine contraction

(4) Milk production

69) In a person, level of PTH is low then the level of which ion is low in blood ?

(1) Iron

(2) Magnesium

(3) Calcium

(4) Chloride

70) Given below are two statements :-
Statement-I : The thymus gland is a lobular structure located between lungs behind sternum on the ventral side of aorta. The thymus plays a major role in the development of the immune system.
Statement-II : Thymus secretes the peptide hormone called thymosin play a major role in the differentiation of T-lymphocytes, which provide cell mediated immunity.
In the light of above statements choose the most appropriate answer from the following.

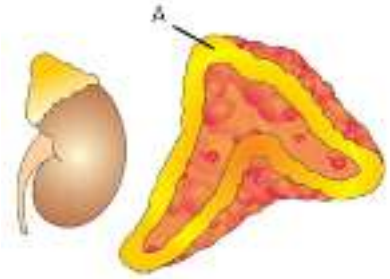
(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

71) Which hormones released from marked area in following diagram ?



(1) Aldosterone, Cortisol and Sex hormone
(2) Aldosterone, Adrenaline and Nor-Adrenaline
(3) Cortisol, Aldosterone and epinephrine
(4) Epinephrine, Nor-epinephrine & Catecholamine

72) Which disorder may be occurred in female after menopause ?

(1) Tetany
(2) Osteoporosis
(3) Rheumatoid arthritis
(4) Myasthenia gravis

73) Name the molecule responsible for the detachment of myosin from actin during mechanism of muscle contraction.

(1) ADP
(2) ATP
(3) CP (Creatine phosphate)
(4) GTP

74) Which structure prevents the embryo from ectopic pregnancy?

(1) Corona radiata
(2) Zona pellucida
(3) Corpus luteum
(4) Graafian follicle

75) Secretion of which gland destroys the acidic medium of male urethra before release of semen ?

(1) Seminal vesicle
(2) Cowpers gland
(3) prostrate gland
(4) Pituitary gland

76) Which of the following statements are correct?
(a) Fertilisation can occur only if the ovum and sperm are transported simultaneously to the ampulla.
(b) By the end of the second month of pregnancy the first movements of foetus can be observed.
(c) On the 14th day of menstrual cycle, ovulation takes place.

(d) Lack of menstruation is always an indication of pregnancy.
Options:

(1) a and c

(2) b and c

(3) c and d

(4) a and b

77) Which of the following statement is incorrect?

(1) The reproductive cycle in the female primates (e.g. monkeys, apes and human beings) is called menstrual cycle.

(2) The menstrual phase is succeeded by the follicular phase.

(3) Maintenance of hygiene and sanitation during menstruation is very important.

(4) Rupture of Graffian follicle occurs 14 days before the beginning of next menstrual cycle.

78) Arrange the components of mammary gland (from proximal to distal)
(A) Mammary Alveoli
(B) Mammary Ampulla
(C) Mammary tubules
(D) Mammary duct
(E) Lactiferous duct

(1) A, C, D, B, E

(2) A, B, E, C, D

(3) D, B, C, A, E

(4) B, E, D, C, A

79) Match the following :

A	Non- medicated IUDs	a	Multiload 375	i	Uterus unsuitable for implantation
B	Copper Releasing IUDs	b	Saheli	ii	Phagocytosis of sperm
C	Hormone Releasing IUDs	c	Lippes loop	iii	Suppress sperm motility
D	Weekly oral pill	d	LNG-20	iv	Anti-estrogenic activity

(1) A-b-i, B-c-ii, C-d-iii, D-a-iv

(2) A-c-iv, B-a-iii, C-b-i, D-d-ii

(3) A-c-ii, B-a-iii, C-d-i, D-b-iv

(4) A-a-iii, B-c-ii, C-d-i, D-b-iv

80) **Assertion (A) :** A healthy person acquires infection of ascariasis through contaminated water, vegetables, fruits etc.
Reason (R) : The eggs of parasite are excreted along with faecal of infected person which contaminate soil, water, plants etc.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

81) A patient is bitten by cobra snake is immediately given injection of preformed antibodies against venom. This is example of

(1) Active immunity

(2) Passive immunity

(3) Innate immunity

(4) Cell Mediated

82) Which of the following acts as cytokine barrier in innate immunity?

(1) Natural killer cell

(2) Skin and mucous membrane

(3) Interferon

(4) Neutrophils

83) Read the following statement and select the option which include all the correct statements only-
(a) Heroin is commonly called as smack.
(b) Heroin is obtained by the acetylation of morphine.
(c) Heroin is a stimulant and increase body functions.
(d) Cocain is a depressant and slows down body functions.
(e) Morphine is a very effective sedative and painkiller.

(1) a, b, c, d

(2) a, c, d, e

(3) a, b, e

(4) b, c, d, e

84) Which of the following is an example of convergent evolution ?

(1) Tiger cat - Banded anteater

(2) Kangaroo - Wombat

(3) Koala - Bandicoot

(4) Mole - Marsupial mole

85) **Assertion :-** Ori, sequence from where replication starts, is important feature of a cloning vector.
Reason :- Piece of DNA when linked to this sequence can be made to replicate within the host cells.

(1) Both assertion and reason are true but reason is not correct explanation of assertion.

(2) Assertion is true but reason is false.

(3) Both assertion and reason are true and reason is correct explanation of assertion.

(4) Assertion is false but reason is true.

86) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as

Reason (R):
Assertion (A) : Foreign DNA and Vector DNA are cut with the help of DNA ligase.
Reason (R) : Ligase acts by forming phosphodiester bonds.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the Correct Explanation of A.

(2) Both A and R are true and R is NOT the Correct Explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

87) Given below are two statements :-
Statement I :- The choice of Bt-genes depends upon the targeted pest only.
Statement II :- The protein encoded by the genes cry I Ab controls corn borer.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true

88) **Statement-I :** The main challange for production of insulin using r-DNA techniques was getting insulin assembeled into a mature form.
Statement-II : Chains A & B were produced separately extracted and combined by creating disulfide bonds to form human insulin.

(1) Both statement-I and statement-II are incorrect

(2) Statement-I is correct but statement-II is incorrect

(3) Both statement-I and statement-II are correct

(4) Statement-I is incorrect but statement-II is correct

89) Correctly match column-I, II and III :-

	Column-I		**Column-II**		**Column-III**
(A)	I^{st} restriction enzyme	(P)	Agarose matrix	(i)	PvuI
(B)	Gel electrophorasis	(Q)	DNA Primers	(ii)	6 base pairs sequence
(C)	pBR322	(R)	Hind II	(iii)	Sieving effect
(D)	PCR	(S)	amp^{R}	(iv)	Amplification of segment of DNA

(1) A-R-ii, B-P-iii, C-S-i, D-Q-iv

(2) A-R-iii, B-P-ii, C-S-i, D-Q-iv

(3) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(4) A-R-ii, B-P-iii, C-Q-i, D-S-iv

90) Which of the following statements are incorrect ?

(A) 'Toddy' a traditional drink of some parts of Southern India is made by fermenting sap from bamboo.
(B) Yeast is used for commercial production of Methanol.
(C) A biological control being developed for use in the treatment of plant disease in the fungus *Trichoderma*.
(D) Many members of the genus *Glomus* from mycorrhiza.

(1) A and B

(2) B and C

(3) B and D

(4) A and C

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	3	4	3	3	1	1	3	2	1	2	3	3	4	3	1	3	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	3	1	1	1	3	2	2	1	4	2	3	2	1	1	4	1	1
Q.	41	42	43	44	45															
A.	1	3	4	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	1	3	4	3	1	3	1	3	3	3	3	1	3	1	2	1	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	1	4	4	2	4	3	1	1	1	3	3	2	4	4	4	2	4	3	3
Q.	86	87	88	89	90															
A.	4	1	4	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	2	4	4	2	1	1	4	4	3	2	4	3	1	2	4	4	1	1	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	1	2	3	4	3	1	3	2	1	3	3	4	3	1	1	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	3	1	2	3	3	3	2	1	1	3	4	3	4	2	2	2	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	1	4	4	2	3	3	3	1	1	2	2	2	2	1	2	1	3	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	3	4	3	4	4	3	1	1										

SOLUTIONS

PHYSICS

1) Potential energy of a dipole in field is

$U = -\vec{p} \cdot \vec{E}$

• Initial dipole moment : $\vec{p}_i = \left(3\hat{i} + 4\hat{j}\right) \times 10^{-30} C \cdot m$

• Field : $\vec{E} = 4000\,\hat{i}\,N/C$

• Dot product : $\vec{p}_1 \cdot \vec{E} = (3 \times 10^{-30})(4000) = 1.2 \times 10^{-26}$

So, $U_i = -1.2 \times 10^{-26}$ J

• Final dipole moment : $\vec{p}_f = \left(-4\hat{i} + 3\hat{j}\right) \times 10^{-30} C \cdot m$

Dot product : $\vec{p}_f \cdot \vec{E} = \left(-4 \times 10^{30}\right)(1000) = -1.6 \times 10^{-26}$

So, $U_f = -(-1.6 \times 10^{-26}) = +1.6 \times 10^{-26}$ J

• Work done by external agent :

$W = U_f - U_i = (1.6 \times 10^{-26}) - (-1.2 \times 10^{-26}) = 2.8 \times 10^{-26}$ J

2)

1. $Z = \sqrt{R^2 + (X_L - X_C)^2}$
2. $I_{rms} = \frac{V_{rms}}{Z}$
3. $P_{coil} = I_{rms}^2 R_{coil}$

$\text{Firstly } Z = \sqrt{(44+36)^2 + (90-30)^2}$

$Z = \sqrt{(80)^2 + (60)^2} = 100\Omega$

So $I_{rms} = \frac{V_{rms}}{Z} = \frac{200}{100} = 2\text{amp}$.

Now $P_{coil} = I_{rms}^2 R_{coil} = (2)^2 (36) = 144\,\text{watt}$

$P_{colil} = 144$ watt

3)

Photon’s energy and momentum are related by

$E = pc$

Here E → Energy, p → momentum, c → Speed of light

 $E = 2 \times 10^{-16} \times 3 \times 10^8 \times 10^2$ erg

$= 6 \times 10^{-6}$ erg

4)

$m = m_0 \times m_e$

$-20 = m_0 \times 5$

$m_0 = -4$

$m_e = \frac{D}{f_e}$ $\quad v^0 + f_e = 14$

$5 = \frac{20}{f_e}$ $\quad v^0 + 4 = 14$

$f_e = 4$ cm $\quad v_0 = 10$

$m_0 = \frac{f_0 - v_0}{f_0}$

$-4 = \frac{f_0 - 10}{f_0}$

$-4f_0 = f_0 - 10$

$f_0 = 2$cm

5)

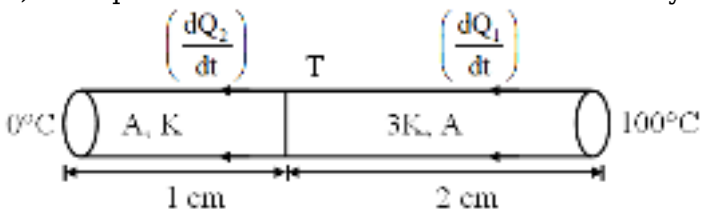
AB → isochoric process ⇒ $W_{AB} = 0$

BC → isothermal process ⇒ $W_{BC} = RT_2 \ln\left(\frac{V_2}{V_1}\right)$

CA → isobaric process (as V ∝ T)

⇒ $W_{CA} = PdV = R(T_1 - T_2)$

6) This question is based on heat current in steady state.



In steady state

$H_1 = H_2$

$\frac{dQ_1}{dt} = \frac{dQ_2}{dt}$

$\frac{3KA(100 - T)}{2} = \frac{KA(T - 0)}{1}$

$300 - 3T = 2T$

$5T = 300$

$T = \frac{300}{5} = 60°C$

7) $C' = \frac{\epsilon_0 A}{d - t + \frac{t}{\epsilon_r}} = \frac{\epsilon_0 A}{\left(d - \frac{2d}{3}\right) + \frac{2d}{3\epsilon_r}} = \frac{7}{6}\frac{\epsilon_0 A}{d}$

on solving

$\Rightarrow \epsilon_r = \frac{14}{11}$

8)

$\Delta E = 13.6\ z^2\left(1-\frac{1}{3^2}\right)$
$z = 3$
$\Delta E = 13.6 \times 8$
$\Delta E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{\Delta E}$
$\lambda = \frac{1240\ eV-nm}{13.6 \times 8\ eV}$
$\lambda = 11.4$ nm

9) Let volume of cube = $V = (10\ cm)^3$
For cube : $W_{app} = W_{act} - F_B$
$W_{app} = V\rho_c g - \frac{V}{2}\rho_w g = V_g(\rho c - \frac{\rho_w}{2})$
$= (10\ cm)^3 g\ (\rho_c - \frac{1}{2})$
From principle of moments :
$W_{app} \times (30 - 10) = (80 - 30) \times 400g$
$(10cm)^3 g(\rho_c - \frac{1}{2})(20) = 50 \times 400\ g$
$10^3(\rho_c - \frac{1}{2}) \times 20 = 20000$
$\Rightarrow \rho_c = 1 + \frac{1}{2} = 1.5\ g/cm^3$
mass of cube $= 1.5 \times (10)^3 = 1500$ g

10) Resistance exhibits non-polar behaviours with V-I graph as straight line ; A→S
Capacitor initially conducts d-c. but stops conducting when fully charged behaving like open circuit- B-R,
Diode/LED shows open circuit in reverse bias but shows fixed voltage drop in forward bias ; C → P
With unregulated voltage, the zener current increases while voltage drop across it, almost remains constant D→Q.

11) Initially $\frac{R}{X} = \frac{75}{25} = 3 \Rightarrow R = 3X$
Later $X_{eq} = \frac{12X}{12+X}$
$\Rightarrow \frac{R}{12X}(12+X) = \frac{80}{20} = 4$
$\frac{3X}{12X}(12+X) = 4 \Rightarrow X = 4\Omega$

12)

Fringe visibility
$= \frac{2\sqrt{\beta}}{1+\beta}$

Where $\beta = \frac{I_1}{I_2} = \frac{1}{4}$

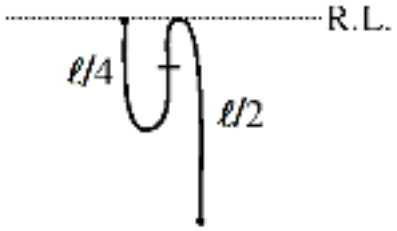
Then

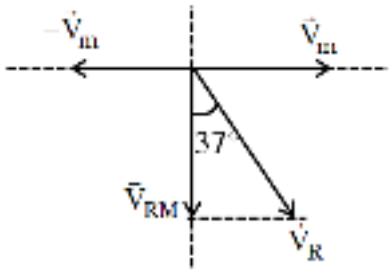
F.V. = 0.8

13)

$$W = -\left[\frac{m}{4}g\frac{\ell}{8} \times 2 + \frac{m}{2}g\frac{\ell}{4}\right] - \left(-mg\frac{\ell}{2}\right)$$

$$W = +\frac{5mg\ell}{16}$$





14)

$\vec{V}_R$ = Velocity of rain = ?

$\vec{V}_m$ = Velocity of man = 72 cm/s

$\vec{V}_{RM}$ = Velocity of rain w.r.t man

From figure.

$$\sin(37^\circ) = \frac{|\vec{V}_m|}{|\vec{V}_R|}$$

$$\frac{3}{5} = \frac{72}{|\vec{V}_R|} \Rightarrow |\vec{V}_R| = \frac{5 \times 72}{3} = 120\,cm/s$$

15)

f_1 = 512Hz; f_2 = ?

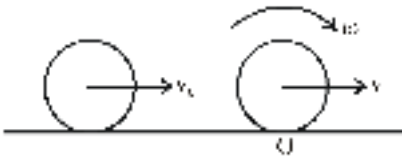
4 = 512 – f_2 or 4 = f_2 – 512

⇒ f_2 = 508 Hz or f_2 = 516 Hz

After increasing tension of piano string f_2 ↑ (i.e. $f_2' > f_2$)

2 = 512 – f_2' Possible

2 = f_2' – 512 Not possible

16) 

By conserving angular momentum about 'O'

$$mv_0R = \left(\frac{2}{5}mR^2 + mR^2\right)\frac{v}{R}$$

$$mv_0R = \frac{7}{5}mR^2\frac{v}{R}$$

$$\Rightarrow v = \frac{5v_0}{7}$$

17) $\vec{I} = \frac{\vec{B}}{\mu_0} - \vec{H}$

$= \frac{1.25}{4\pi} \times 10^7 - 5 \times 10^5 = 5 \times 10^5$ A/m.

18) $\frac{T_{max}}{T_{min}} = \frac{3}{2}$ in vertical circular motion and asking about speed at lowest point.

Solution :

$T_{max} - T_{min} = 6mg$ ....(i)

$$\frac{T_{max}}{T_{min}} = 1.5 = \frac{3}{2}$$

$$T_{min} = \frac{2}{3}T_{max} \quad \text{...(ii)}$$

Using (i) and (ii), we get :-

$T_{max} = 18$ mg

At lowest point :- $T_{max} - mg = \frac{mv^2}{\ell}$

$\Rightarrow v = \sqrt{17g\ell}$

19) $v_{rel}\frac{dm}{dt} - mg = ma$

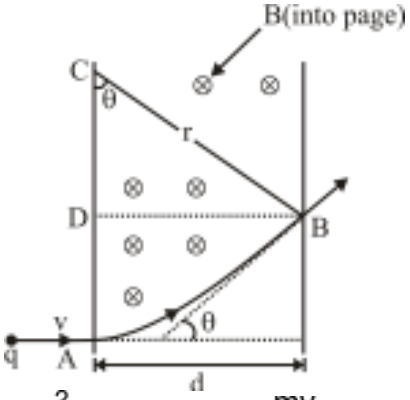
$$\Rightarrow \frac{dm}{dt} = \frac{(mg + ma)}{v_{rel}}$$

20)

Let v be the velocity of the particle.Its kinetic energy is

$\frac{1}{2}mv^2 = qV$ or $v = \left(\frac{2qV}{m}\right)^{1/2}$ (1)

The particle follows a circular path from A to B of radius r which is given by



$\frac{mv^2}{r} = qvB$ or $r = \frac{mv}{qB}$

Using (1) and (2), we have

$r = \frac{m}{qB}\left(\frac{2qV}{m}\right)^{1/2} = \frac{1}{B}\left(\frac{2mV}{q}\right)^{1/2}$

In triangle BCD, $\sin\theta = \frac{BD}{BC} = \frac{d}{r}$. Therefore,

$\sin\theta = Bd\left(\frac{q}{2mV}\right)^{1/2}$

21) Time taken from A to B is 2 second time to go upto maximum height from A is 1 second. Maximum height is 5m above A. So maximum height is 15 + 5 = 20 m

22) True value of multiplication

⇒ C = AB = (2.5 m/s)(0.10 sec) = 0.25m

Percentage error in multiplication

$\Rightarrow \left(\frac{\Delta C}{C} \times 100\%\right) = \left(\frac{\Delta A}{A} \times 100\%\right) + \left(\frac{\Delta B}{B} \times 100\%\right) = \left(\frac{0.5}{2.5} \times 100\%\right) + \left(\frac{0.01}{0.10} \times 100\%\right)$

$= (20\%) + (10\%)$ $\left(\frac{\Delta C}{C} \times 100\%\right) = 30\%$ Absolute error in multiplication ⇒ ΔC = 30% of C = $\frac{30}{100} \times 0.25m = 0.075$ m ≈ 0.08m So, error full value of multiplication ⇒ C = 0.25 m ± 0.08 m

23)

Intensity $I = \frac{1}{2}\in_0 E_0^2 C$

$\Rightarrow E^0 = \sqrt{\frac{2I}{\in_0 C}}$

$= \sqrt{\frac{2 \times 3.3}{8.8 \times 10^{-12} \times 3 \times 10^8}}$

= 50 N/C

and $B^0 = \frac{E_0}{c} = \frac{50}{3 \times 10^8} = 1.67 \times 10^{-7}$ T

24) $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2gR^2}{R}} = \sqrt{2gR} = \sqrt{2 \times \frac{4}{3}\pi G\rho R \times R}$

$= \sqrt{\frac{8}{3}\pi G\rho R^2}$

$= 2R\sqrt{\frac{2}{3}\pi G\rho}$

$\Rightarrow V_e \propto R\sqrt{\rho}$

$\because \rho$ = constant

$v_e \propto R$

$\Rightarrow \left(\frac{\Delta v_e}{v_e} \times 100\%\right) = \left(\frac{\Delta R}{R} \times 100\%\right)$

= –4 %

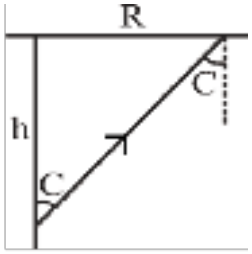
= decrease by 4%

25) $\frac{\sin 90^\circ}{\sin C} = \mu$

$\sin C = \frac{1}{\mu} \Rightarrow \frac{R}{\sqrt{R^2 + h^2}} = \frac{3}{4}$

Squaring, $16R^2 = 9R^2 + 9h^2$

$7R^2 = 9h^2 \Rightarrow R = \frac{3}{\sqrt{7}}h.$



26) We have to find the heat supplied to the gas.

One mole of ideal monatomic gas at T^0 expands slowly under law $\frac{P}{V}$ = constant.

Final temperature is $2T_0$.

Thermodynamics: molar heat capacity for a polytropic process PV^x = constant, First law of thermodynamics.

Given: $\frac{P}{V}$ = constant, so PV^{-1} = constant $\Rightarrow$ x = – 1

For monatomic gas: $\gamma = \frac{5}{3}$

Molar heat capacity in polytropic process: $C = C_v + \frac{R}{1-x}$ Here $C_V = \frac{3}{2}$ R for monatomic gas.

$C = \frac{3}{2}R + \frac{R}{1-(-1)}$ $C = \frac{3}{2}R + \frac{R}{2} = 2R$ Temperature change $\Delta T = 2T^0 - T^0 = T^0$

Heat supplied $\Delta Q = nC\Delta T = 1 \times 2R \times T_0 = 2RT_0$.

27) Time $\propto c^x G^y h^z \Rightarrow T = kc^x G^y h^z$

Putting the dimensions in the above relation

$\Rightarrow \quad [M^0L^0T^1] = [LT^{-1}]^x[M^{-1}L^3T^{-2}]^y[ML^2T^{-1}]^z$

$\Rightarrow \quad [M^0L^0T^1] = [M^{-y+z}L^{x+3y+2z}T^{-x-2y-z}]$

Comparing the powers of M, L and T

$-y + z = 0$ ...(i)

$x + 3y + 2z = 0$ ...(ii)

$-x - 2y - z = 1$ ...(iii)

On solving equations (i) and (ii) and (iii)

$$x = \frac{-5}{2}, y = z = \frac{1}{2}$$

Hence dimension of time are $[G^{1/2}h^{1/2}c^{-5/2}]$

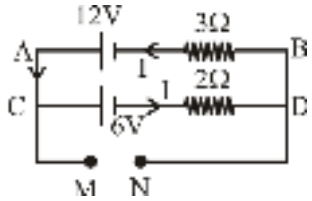
28) Torque,

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 3 \\ 2 & -3 & 4 \end{vmatrix}$$

$= \hat{i}(8+9) - \hat{j}(12-6) + \hat{k}(-9-4)$

$= 17\hat{i} - 6\hat{j} - 13\hat{k}.$

29)



$$I = \frac{12-6}{3+2} = \frac{6}{5}A$$

$$V_{MN} = 6 + 2I = 6 + \frac{12}{5}$$

$$V_{MN} = \frac{42}{5} = 8.4V$$

30) First particle is comming towards mean position

$\therefore \phi_1 = \pi - \frac{\pi}{6}$

second particle is comming towards mean position but in negative side

$\therefore \phi_2 = 2\pi - \frac{\pi}{6}$

$\phi_2 - \phi_1 = \left(2\pi - \frac{\pi}{6}\right) - \left(\pi - \frac{\pi}{6}\right) = \pi$

31)

pitch = $\frac{2mm}{4}$ = 0.5 mm ; LC = $\frac{0.5mm}{100}$ = 0.005 mm

ZE ⇒ Positive

MSR = 0th div = 0 mm

CSR = + 6 × 0.005 = + 0.030 mm

ZE = + 0.030 mm

During measurement

MSR = 4 dir = 4 × 0.5 mm = 2 mm

CSR = 23 × 0.005 = 0.115 mm
CR = 2mm + 0.115 mm – 0.030 mm = 2.085 mm

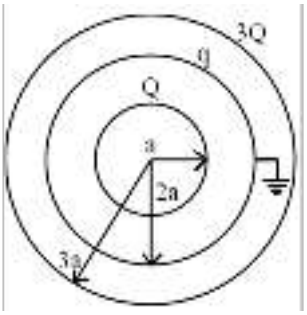
32) $G = \frac{RS}{R-S} = \frac{(4900)(98)}{4900-98} = 100\,\Omega$

Full scale current $I_g = \frac{E}{R+G}$

$= \frac{5}{4900+100} = 1mA$

F.O.M. $= \frac{I_g}{n} = \frac{1mA}{20} = 50\ \mu A/div.$

33)



$\frac{KQ}{2a} + \frac{Kq}{2a} + \frac{K(3Q)}{3a} = 0$

$\frac{q}{2} = -\frac{3Q}{2}$

$q = -3Q$

$q_{flow} = q_i - q_f$

$= -2Q - (-3Q)$

$= Q$

34)

Initially $h\upsilon = \phi + k_{max}$ ..... (1)

$\frac{v^1 - v}{v} \times 100 = 25 \Rightarrow v^1 = \frac{5}{4}v$

Finally $h\left(\frac{5}{4}v\right) = \phi + 2k_{max}$ ..... (2)

$\frac{(1)}{(2)} \Rightarrow \frac{4}{5} = \frac{\phi + k_{max}}{\phi + 2k_{max}}$

$4\phi + 8\ k_{max} = 5\phi + 5\ k_{max}$

$\Rightarrow \phi = 3\ k_{max}$

$= 3 \times 1.8$ eV

$= 5.4$ eV

35) $R_f = \frac{\Delta V}{\Delta I} = \frac{0.8-0.7}{(16-8)} \times 10^3 = 12.5\ \Omega$

$R_r = \frac{15}{1.5} \times 10^6 = 10^7\ \Omega$

$$\frac{R_r}{R_f} = \frac{10^7}{12.5} = 8 \times 10^5$$

36)

n = electron density = $\frac{N}{V} = \frac{N}{A\ell}$

No. of moles = $\mu = \frac{N}{N_A} = \frac{M}{M_\omega} = \frac{M/V}{M_\omega/V} = \frac{dV}{M_\omega}$

$n = \frac{N}{V} = \frac{N_A d}{M_\omega}$

$\because i = neA\, V_d$ hence $V_d = \frac{i}{neA} = \frac{iM_\omega}{N_A deA}$

$Vd = \frac{16(60 \times 10^{-3})}{6 \times 10^{23} \times 5 \times 10^3 \times 1.6 \times 10^{-19} \times 10^{-6}}$

$V_d = 2 \times 10^{-3}$ m/s = 2mm/sec

37)

$F = 2T\ell$

or $\rho A\ell g = 2T\ell$

or $\rho \frac{\pi d^2}{4} g = 2T$

or $d^2 = \frac{8T}{\rho\pi g}$

$d^2 = \frac{8 \times 22.5\pi \times 10^{-3}}{8 \times 10^3 \pi \times 10}$

$d^2 = 2.25 \times 10^{-3}$

or d = 1.5 mm

38) For k to be negative, angle of contact for one liquid has to be acute (cosθ = + ve) and for another liquid it should be obtuse (cosθ ⇒ –ve).

39)

425 : 595 : 765

5 : 7 : 9

ratio of frequency is odd

pipe → closed end.

425 is frequency of 2nd overtone

$\frac{5v}{4\ell} = 425 \Rightarrow \frac{5 \times 340}{4\ell} = 425$, $\ell = 1$m

40) 

1.5 m/s

By COLM
$mv_1 + 2mv_2 = m \times 1.5$
$v_1 + 2v_2 = 1.5$ .....(i)
By newton's law of collision
$v_2 - v_1 = 0.9$ .....(ii)
By (i) & (ii)
$v_2 = 0.8$ m/s
& $v_1 = -0.1$ m/s

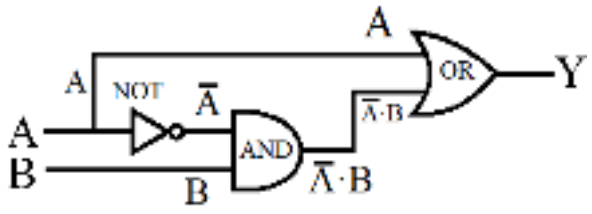
41)

Magnetic field at the end of a solenoid
$B = \left(\frac{1}{2}\right)\mu^0 nI$
where $n = \frac{N}{L}$ = Number of turns per unit length
Given:
L = 50cm = 0.5 m
N = 100
I = 2.5A
n = N/L = 100/0.5 = 200 turns per meter
$B = \left(\frac{1}{2}\right) \times (4\pi \times 10^{-7})\ (200)\ (2.5)$
$B = \left(\frac{1}{2}\right) 2000\pi \times 10^{-7}$
$B = 1000\pi \times 10^{-7}$
$B = \pi \times 10^{-7} T$
$B \approx 3.14 \times 10^{-4}$ T

42)

The mass of the fuel is 2000g. Since the molar mass of deuterium ($_1H^2$) is 2 g/mol, the number of nuclei N is :
$$N = \frac{Mass}{MolarMass} \times N_A$$
$$= \frac{2000}{2} \times 6.022 \times 10^{23}$$
$$= 6.022 \times 10^{26} nuclei$$
The question show that 2 deuterium nuclei are consumed per reaction. Thus, the total number of reactions n is:
$$n = \frac{N}{2} = \frac{6.022 \times 10^{26}}{2} = 3.011 \times 10^{26}$$ reactions
Each reaction releases 3.2 MeV To convert this to electron volts (eV), we are 1 MeV = 10^6 eV :
$E = n \times 3.2 \times 10^6$ eV = $3.011 \times 10^{26} \times 3.2 \times 10^6 = 9.6352 \times 10^{32}$ eV
$\approx 10^{27}$ MeV

43)



$Y = A + \overline{A} \cdot B$

Using the rule Y = A + BC = (A+B) (A+C)

$Y = \left(A + \overline{A}\right) \cdot (A + B)$

$Y = (1) \cdot (A + B)$

Y = A + B

⇒ OR Gate

The corresponding Truth Table will be-

A	B	$Y(A+B)$
0	0	0
0	1	1
1	0	1
1	1	1

44) $Q = \frac{\Delta\phi}{R} = \frac{NBA}{R}$

so $B = \frac{QR}{NA} = \frac{2 \times 10^{-5} \times 20}{40 \times 0.01} = 1 \times 10^{-3} T$

45) $B = \frac{P}{-\left(\frac{\Delta V}{V}\right)}$

$\Rightarrow 138 \times 10^9 = \frac{345 \times 10^6}{-\left(\frac{\Delta V}{V}\right)}$

$\Rightarrow \frac{\Delta V}{V} = -\frac{345 \times 10^6}{138 \times 10^9} = -2.5 \times 10^{-3}$

$\Rightarrow \% \frac{\Delta V}{V} = -2.5 \times 10^{-3} \times 100 = -0.25\ \%$

'Negative sign' indicates that volume decreases, if pressure is increased and vice-versa.

CHEMISTRY

46)

(i) 1s and 2s are spherical → Correct
(ii) Probability is not minimum near nucleus → Incorrect
(iii) Probability at a given distance is equal in all directions → Correct
(iv) Probability density of 2s decreases uniformly → Incorrect

47)

$H_2 + I_2 \rightleftharpoons 2\,HI$

ni	1	1	0
n_{eq}	0.2	0.2	1.6

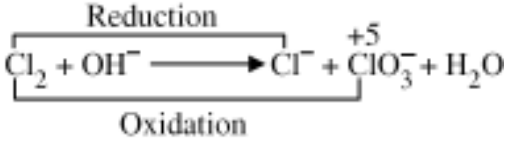
$$K_c = \frac{(1.6)^2}{0.2 \times 0.2} = 64$$

48)

$Ksp(Ag_2CO_3) = 4 \times 10^{-13}$
$[CO_3^{2-}] = 0.1$ M
Let solubility of $Ag_2CO_3 = s$
$Ksp = [Ag^+]^2 \times [CO_3^{2-}]$
$4 \times 10^{-13} = (2s)^2 \times 0.1 \approx 4s^2 \times 0.1 = 0.4\ s^2$
$s^2 = 4 \times 10^{-13} / 0.4 = 1 \times 10^{-12}$
$s = 10^{-6}$ M

49) $2 \times 10^{-3} \times N_A \times 10$

50) $N_1V_1 = N_2V_2$
$0.1 \times 5 \times 10 = M_2 \times 1 \times V_2$
$M_2V_2 = 5$
(Fe^{+2})

51) 

52) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$
$\Delta H = \Delta H_P - \Delta H_R$

53)

(A) When liquid evaporates to vapour, randomness increases hence entropy increases.
(B) Temperature decreases ⇒ entropy decreases
(C) $\Delta n_g = 2 - 0 = 2$
$\Delta n_g = +$ ve $\rightarrow \Delta S = +$ ve
⇒ entropy increases
(D) $\Delta n_g = 2 - 1 = 1$
$\Delta n_g = +$ve $\rightarrow \Delta S = +$ ve
⇒ entropy increases
(A), (C), (D) are correct.
Final answer : 3

54) R.O.R = $K[A_2]^a[B_2]^b$

$0.04 = K(0.2)^a(0.2)^b$ ... (i)

$0.04 = K(0.1)^a(0.4)^b$ ... (ii)

$0.08 = K(0.2)^a(0.4)^b$ ... (iii)

$\frac{(i)}{(iii)}\ \frac{1}{2} = \left(\frac{1}{2}\right)^b$; $b = 1$

$\frac{(i)}{(ii)}\ \frac{1}{1} = 2^a \times \left(\frac{1}{2}\right)^b$; $a = 1$

55)

Z_{AB} = Collision frequency of reactants A and B

$e^{-Ea/RT}$ = Fraction of molecules with energies equal to or greater than activation energy (E_a)

56)

During electrolysis of molten sodium hydride (NaH), Na^+ is reduced at the cathode and H^- is oxidized at the anode, releasing H_2 gas.In electrolytic refining, impure metal is made the **anode**, and pure metal is deposited at the **cathode**.

Statement I: Correct → H^- ions lose electrons at the anode to form H_2 gas.

Statement II: Incorrect → Pure metal is obtained at the cathode, not the anode.

57) $E = 0.323 - \frac{0.059}{2} \log \frac{10^{-3}}{5 \times 10^{-3}}$

58) $0.0821 = 1 \times C \times 0.0821 \times 300$

$C = \frac{1}{300} \Rightarrow C = 0.3 \times 10^{-2} \frac{mol}{L}$

59)

$T_f° - T_f = k_f \times m$

$0 - (-0.325) = 1.86 \times m$ .....(1)

$T_b - T_b° = K_b \times m$

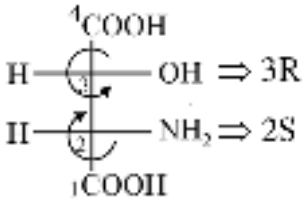
$T_b - 100 = 0.52 \times m$ .....(2)

$\frac{0.325}{T_b - 100} = \frac{1.86}{0.52}$

$T_b - 100 = \frac{0.325 \times 0.52}{1.86} = 0.090\ °C$

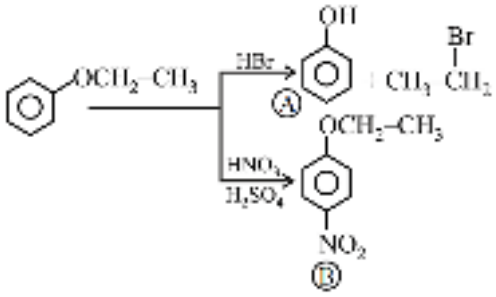
$T_b = 100.09\ °C$

60) During reaction of strong acid and strong base 57.2 kJ eq^{-1} heat is released.

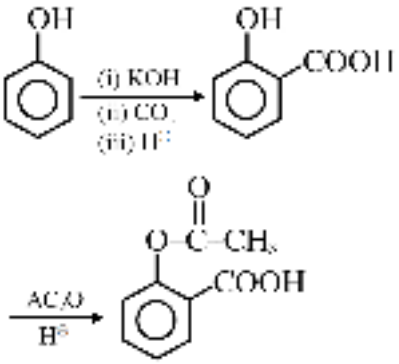


61)

(2S, 3R)-2-Amino-3-hydroxy butane-1, 4-dioic acid



62)



Acetyl salicyclic acid
(Aspirin)

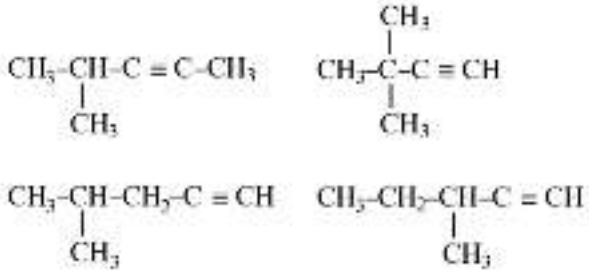
63)

64)

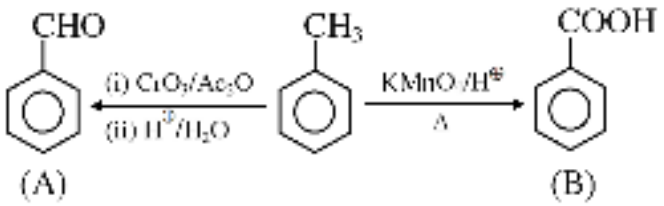
C_6H_{10} $CH \equiv C–CH_2–CH_2–CH_2–CH_3$

$CH_3–C \equiv C–CH_2–CH_2–CH_3$

$CH_3–CH_2–C \equiv C–CH_2–CH_3$



Total structural isomer 7

65) 

66)

Aromatic aldehyde does not react with fehling solution.

67)

% enol ↑ When enol formed is
(i) Aromatic
(ii) Extended reso
(iii) H-bonding

68)

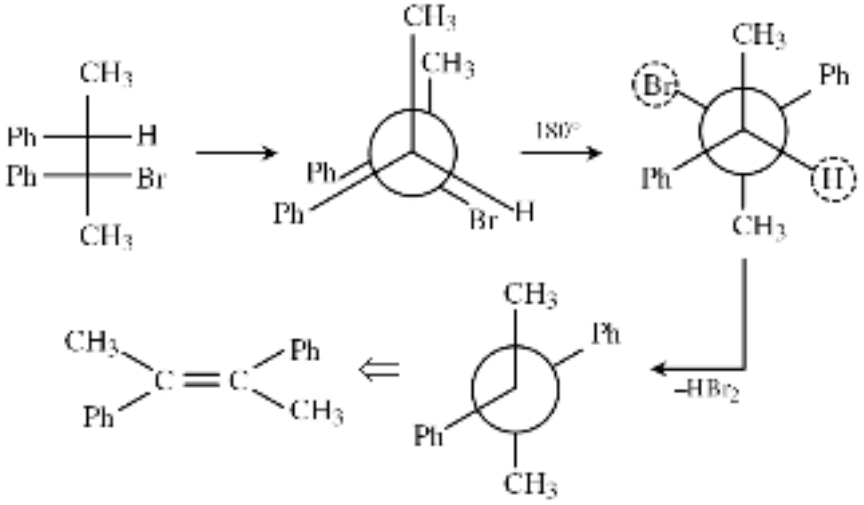
Rate of $SN^2 \propto \frac{1}{\text{steric hindrance}}$

69)

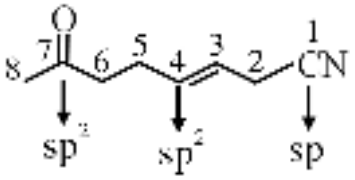
NCERT-XII Pg. # 394 & 395

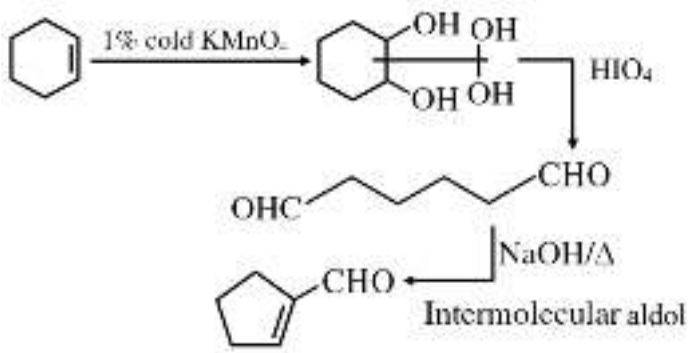
70)

In E-2 reaction β-hydrogen and leaving group should be anti



Final Answer :- (2)

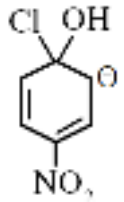
71) 

72) 

73)

NCERT XII Pg. # 410

74) $C_xH_yN_z + CuO \rightarrow CO_2 + H_2O + N_2 + Cu$

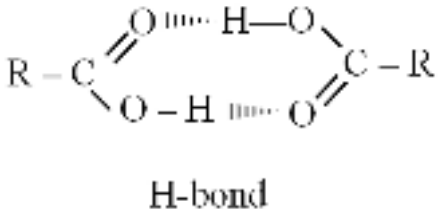
75) 

Stabilised by –M effect of $-NO_2$

76) NCERT-XI Pg. # 91
B < Li < C < O < F
We know EA order

77) $Na^+ > Mg^{+2}$ (iso e^-)
$C\square^{35} \approx C\square^{37}$ → Isotope
 same size

78) C_2 exist in vapour phase at high temperature [NCERT]

79) 

H-bond

80)

NCERT Pg. no. # 112, 107
For greater covalent character of an ionic bond (i) small cation (ii) large anion.

81)

NCERT Pg. # 107

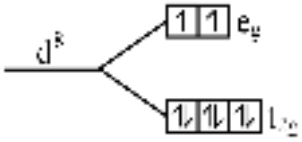
82) For C.N = 6 O.S of Co → +3

$[CoF_6]^x$ $K^+ + [CoF_6]^{3-}$

↓ ↓

3 – 6 = x $K_3[CoF_6]$

x = –3



83) $Ni^{2+} \rightarrow [Ar]3d^8$

$t_{2g}^6\ eg^2$

CFSE = $(-0.4 \times nt_{2g} + 0.6 \times n_{eg})\Delta_0$

= $(-0.4 \times 6 + 0.6 \times 2)\Delta_0$

= $(-2.4 + 1.2)\Delta_0$

⇒ $-1.2\ \Delta_0$

84) $NiCl_2.6H_2O \rightarrow [Ni(H_2O)_6]Cl_2$

$\xrightarrow[AgNO_3]{excess}$ 2 moles of AgCl Precipitated

85) Cu^{2+} is more stable because more –Ve $\Delta_{hy}H$ of Cu^{+2}, compensate second ionisation enthalpy of Cu.

86) All of these

87) NCERT Pg. # 220

88) NCERT Pg. # 181

89) NCERT-XII, Pg. # 101
Bluish - green

90)

The common precipitating reagent for Group II is hydrogen sulphide (H_2S) in the presence of

dilute HCl. They precipitate as **sulphides**, not hydroxides. Group III radicals (like Fe^{3+}, Al^{3+}, Cr^{3+}) precipitate as hydroxides.

Final Answer :- 4

BIOLOGY

91) NCERT-XI - Page No. 4

92) NCERT-XIth Pg # 31, Fig. 3.1, Module

93) NCERT (XI) Pg. # 30, para-01

94) **Explain Question :** Question asks which statements about Mycoplasma are correct.
Concept : This question is based on Mycoplasma
Explanation : Mycoplasma: smallest living organisms, pleomorphic, lack cell wall.Can survive without oxygen (facultative anaerobes).Many species are pathogenic to plants and animals.All four statements (I–IV) are correct.
Final answer : 4

95) NCERT-XII Pg # 11

96)

NCERT XI, Pg # 21

97) NCERT Pg. # 67, 68

98)

NCERT-XI, Pg.No.# 74

99)

NCERT XI Pg.# 71,72,75

100)

NCERT XI Pg. # 73

101) NCERT Pg XI # 93,94

102) **Question Explanation :** The question gives 4 matches between organelles and their functions, and asks which one is *correct*.
Concept : Cell Organelles – Functions
Solution: Peroxisome → Involved in oxidation of fatty acids & detoxification, *not* glucose.
Lysosome → Contains hydrolytic enzymes for digestion, *not* lipid synthesis.
Smooth ER (SER) → Synthesizes lipids, steroids, detoxifies drugs, *not* proteins.
Rough ER (RER) → Studded with ribosomes, main site of protein synthesis.

Final Answer: Option (4)

103)

Explain Question : The question asks which organelle-function pair is correct for the given diagram (likely showing a cellular organelle).
Concept : This question is based on Golgi complex
Explanation : Golgi complex – Modifies proteins, forms glycoproteins, sorts and packages them.
Final answer : 3

104) Module # 01 Pg. # 174

105) NCERT XI, Pg. # 126

106)

Ncert page 124.

107) NCERT Pg. # 154

108) Module no. 3, pg.no. 178 (E), 194 (H)

109) NCERT-XI Pg. # 250

110) **Asking about:**
coefficient of mutual inductance

Concept:
Mutual induction

Solution:
$\because \phi = Mi$
or $M = \frac{\Delta\phi}{\Delta i} = \frac{2 \times 10^{-2}}{0.01} H$
or M = 2H
correct option: (1)

111) NCERT-XI Pg. No. # 232

112)

Explain Question: The question asks to initiate flowering and synchronizing the fruit set in pineapple.

Concept: Ethylene.

Solution:

The hormone which is used to initiate flowering and synchronizing the fruit set in pineapple is known as:

Ethylene.

So, the correct answer is: 2.

Final answer: Option 2

113)

Question Explanation : The question is asking what Geitonogamy is ,specifically, its functional and genetic nature.
Concept: Pollination.
Solution: Geitonogamy is transfer of pollen grains from the anther to the stigma of another flower of the same plant. Although geitonogamy is functionally cross-pollination involving a pollinating agent, genetically it is similar to autogamy since the pollen grains come from the same plant.
Final Answer: Option (3)

114) **Explaining :**
Concept :
Solution : NCERT XII Pg. # 31
In monoecious pants since male and female flowers are present on different plant so autogamy is not possible but geitonogamy and xenogamy is possible.
Final Answer : (1)

115) NCERT (XII) Pg. # 26,27(E) 28,29(H)

116) NCERT XII, Pg # 25

117)

101 Module

118)

XIIth New NCERT Pg. # 67

119)

NCERT-XII, Pg. # 55,60,61

120) Both mitochondria and chloroplasts have their own DNA and are responsible for cytoplasmic inheritance, This refers to the inheritance of traits determined by genes found in organelles outside the nucleus, specifically mitochondria and chloroplasts.

121)

Explaining the Question :-The question refers to Morgan's work on dihybrid crosses in Drosophila (fruit flies), where he discovered the concept of linkage-the tendency of genes located close together on the same chromosome to be inherited together. The question asks about his observations regarding recombination and likage.The question asks about his observations regarding recombination and likage.
Concept : -

Solution : Morgan discovered that are tightly linked (located very close to each other on the same chromosome) tend to show low recombination frequencies.

The 1.3% recombination frequency between the white eyes and yellow body genes in Drosophila indicates they are very closely linked, with very few crossover events occurring between them.
Final Answer :- (2)

122)

Question Explanation: Identifying the types of RNA essential for the translation process.

Concept: Roles of different RNA types in Protein Synthesis

Solution:

All three major classes of RNA are necessary for protein synthesis (translation):

A. mRNA (messenger RNA): Serves as the template or blueprint carrying the genetic code from DNA.

B. tRNA (transfer RNA): Functions as the adapter molecule, carrying specific amino acids to the ribosome and matching them to the mRNA codons.

C. rRNA (ribosomal RNA): Is the structural and catalytic component of the ribosome, providing the framework and possessing the peptidyl transferase activity.

hnRNA (heterogeneous nuclear RNA) is the precursor to mRNA in eukaryotes, but it is not directly involved in translation itself.

Final Answer: 1

tRNA, mRNA, rRNA

123) NCERT-XII Pg. No. # 109, Fig- 6.10

124)

Question Explanation: Difference between RNA and DNA

Concept: Transcription

Solution:
RNA differs from the DNA template because it contains uracil (U) instead of thymine (T). Exons and introns are segments transcribed but do not change base identity.

Final Answer: 3

125) **Explain Question:** The question asks about the correct match about the nucleotides.
Concept: Nucleotides
Solution:
(i) ϕ × 174 Bacteriophage : 5386 N-bases
(ii) Lambda bacteriophage : 48502 bp
(iii) E. coli: 4.6 × 106 bp
(iv) Human: 3.3 × 109 bp
So, the correct answer is: 4. (i) = C, (ii) = D, (iii) = A, (iv) = B
Final answer: 4.

126) NCERT-XII Pg # 83, 84, 85

127)

NCERT Page No. # 118

128) NCERT-XI Pg. # 145

129) NCERT XII # 236, 237

130)

NCERT XII Pg # 196

131)

NCERT-XII Pg. # 207

Explaining the Question :-

This asks how many statements about productivity and decomposition are true.

Concept :- productivity and decomposition

Solution :- a) True – NPP = biomass for heterotrophs

b) True – Secondary productivity = new organic matter by heterotrophs

c) True – Warm/moist speeds decomposition

Final Answer :- (3) a, b, d

132) NCERT XII, Pg. # 194, 195

133) NCERT Page No. 245

134) NCERT Page No. 267

135)

Question Explanation:

Identifying the monomers of sucrose.

Concept:

Composition of Disaccharides

Solution:

A. REASON: Sucrose (common table sugar) is a disaccharide, meaning it is formed from two simpler sugar units (monosaccharides).
B. REASON: These two units are one molecule of α-D-glucose and one molecule of β-D-fructose, joined by a glycosidic bond.

Final Answer:

The correct option is 2.

136) NCERT XI Page No.: 49, 50

137) NCERT, Pg. # 45

138) NCERT, Pg. # 49

139) NCERT, Pg. # 38

140)

NCERT XI, Pg # 47, 48

141) NCERT Pg. # 104

142) NCERT Pg. No. # 114

143) NCERT Pg. No. # 113

144) NCERT Pg # 82, 83

145) NCERT-XII, Pg. No. 82

146) NCERT XI, Pg. # 194

147) NCERT Pg # 194

148) NCERT Class-XII, Page No. 195, 196

149) NCERT, Pg. # 187

150) NCERT, Pg. # 191

151) NCERT Pg. No. # 189

152)

NCERT XI, Pg # 213

153) NCERT BASED

154) NCERT (XII) ; Page No. # 321 (E)
NCERT (XII) ; Page No. # 321 (H)

155) NCERT (XII) ; Page No. # 317 (E)
NCERT (XII) ; Page No. # 317 (H)

156) NCERT Pg. # 232

157)

NCERT XI, Pg # 240, 241

158)

NCERT XI, Pg # 241

159) NCERT Pg.#243

160) NCERT Pg.#243

161)

Solution/Explanation:
The image shows a diagram of the adrenal gland, with the **adrenal cortex** (outer layer) and **adrenal medulla** (inner core) highlighted. The question asks which hormones are released from the **marked area**, which is the **adrenal cortex**.

Option 1: Aldosterone, Cortisol, and Sex hormone. These are produced by the adrenal cortex (marked area).

162) NCERT Pg.#227

163) NCERT XI Pg.#222

164)

NCERT XII, Pg # 34

165) NCERT, Pg # 28

166) NCERT(XII) Chapter # 2

167) NCERT XII Pg. No. # 33, 34, 35

168) NCERT, Pg # 30

169) **What the Question Is Asking:**match the types of contraceptives listed in column A. with their specific examples and mechanisms of action in column B.**Solution AND CORRECT ANSWER:**

A. **Non-medicated IUDs:(** Lippes loop) They can cause a mild inflammatory reaction facilitate Phagocytosis of sperm.

B. **Copper Releasing IUDs:(**Multiload 375) Copper ions released by these IUDs suppress sperm motility.
C. **Hormone Releasing IUDs:** (LNG-20) These IUDs release hormones like progestin,.primarily work by making the uterus unsuitable for implantation.

Weekly Oral Pill (Saheli): This pill contains a non-steroidal compound that has anti-estrogenic activity, preventing ovulation. Therefore, the correct matching is:**(3) A-c-ii, B-a-iii, C-d-i, D-b-iv**
Difficulty Level: Medium

170)

NCERT XII, Pg # 130, 131

171) NCERT-XII, Pg. # 136

172) NCERT-XII, Pg. # 135

173)

NCERT PG. # 142, 143

174) NCERT XII Pg. No. # 118

175) NCERT-XII, Pg. # 169

176) NCERT, Pg. # 165

177)

Explain Question : The question asks about the correctness of statements related to Bt genes and their pest targets.

Concept : Genetically Modified Crops – Bt genes

Explanation :
Statement I: “The choice of Bt-genes depends upon the targeted pest only.”

– Not entirely correct; gene choice also depends on crop type, expression level, and ecological safety, not just the pest → False

Statement II: “The protein encoded by cry IAb controls corn borer.”

–cry IAb is effective against European corn borer → True

Final answer : Option (4)

178) NCERT Pg. # 181,182

179) NCERT XII Pg. # 165,168,169,173 (E), 181,184,185,189 (H)

180) NCERT, Pg. # 151,153,157,158